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DRIVERS FOR PURCHASE/REPURCHASE INTENTIONS IN MOBILE GAMES APPLYING THE FREEMIUM BUSINESS MODEL

International Business and Sales
Master's Thesis
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ABSTRACT

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The freemium business model imposed itself as a leading concept in the gaming industry. This success translated into an increase of the overall number of the new F2P games created every year. This rise in popularity in return, led to the realisation of the pungent need to understand what exactly are the reasons that stand behind its prosperity. Couple of studies takes on this task resorting to the theories of consumption and technology acceptance and applying abstract concepts as perceived value, perceived usefulness and service quality. However, none of those studies focus primarily on the free-to-play mobile games. The market statistics indicate that the mobile gaming industry shapes to grow exponentially in the future and more accent has to be placed on deciphering the mobile games' players behaviour. Furthermore, the extant literature lacks knowledge regarding the role that individuality and personality of the players plays in their willingness to spend money. The current research addresses this void in the literature by conducting quantitative analysis on a data sample of 527 respondent that confirmed to be playing mobile games. The built model aims to depict a complete picture of the drivers behind players purchase intentions as a distinction is made between perceived value and overall value of the game as well as perceived service quality

is included. Also, the model was tested for moderation effects on players' purchase intentions that might take place when personal level of cultural orientations and personality traits are taken into consideration. The results of the analysis reveal differences between the group of spenders and non-spenders in terms of factors that predict their sense of overall value of the game as well as their purchase intentions. While spenders give more value to the ability of the game to provoke emotions in them (emotion response) non-spenders appreciate more the level or reliability. Unlike previous researches, in the context of mobile free-to-play games the indicator of assurance provided in the game in terms of safety for personal and bank information, turns out to be important indicator for purchase intentions. In addition, for the non-spenders, negative effect was observed from the reliability of the game towards the purchase intentions variable. This effect was found out to be mitigated by the personality trait extraversion and the cultural orientation social inequality. The analysis of the spenders group confirms previous observation about the importance of monetary price aspect and also discovers interesting dynamics regarding the power orientation of the players and how inclined they are to buy premium content.

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1. INTRODUCTION

1.1. Background of the study

The rapid growth of technologies resulting from the globalization and the new technological era that we are living in, gradually conquered every sphere of our everyday life to the point that now it is hard to even imagine a full day to pass without the help of some sort of technological product. One of the most influential technological discovery which find its way in the life of the young population as well as the old one, is the invention of internet. Internet has proved over the years to bring about many advantages to the modern society and its potential was immediately recognized by the businesses. Arguably, the most prominent aspect of the radical technological development marking its debut in the twenty century is the change of the way of communicating. The less time needed to transfer information from one place to another substantially accelerated the development of new technologies and completely changed social, cultural and business landscapes. Nowadays, people rely heavily on computers when performing all kinds of activities which gradually made computers ubiquitous, and thereby the digital economy grew to become a highly important indicators of the national economic growth and competitiveness (Barefoot,2018). In general, the notion of digital economy comprises the adoption of information and communication technologies (ICTs) designed to perform different activities. Typical example of such implementation of technologies is the shift from brick-and-mortar stores to an e-commence and from standard physical goods to digital ones (Barefoot,2018). To that end, we can distinguish two types of digital economy – one that is using digital technologies to promote the consumption of goods and services, and the other presenting digital technologies as consumption objects (Digital virtual consumption).

This evolution in consumption led to the development of *new sort of goods – virtual ones*. Throughout the years many authors provided diverse definitions of ‘’virtual goods’’ as some take the point of view of the service that is examined determining them as goods existent in a virtual world while other also take into account the e-commerce transaction as MP3s and movies. Yet, among all definitions, Fairfield seems to deliver the best description of virtual goods as he makes distinction between information goods and virtual goods (Lehdovirta, 2009). While the virtual goods are by nature digital contrary to the conventional tangible products, they are nevertheless rivalrous and excludable by design within the environment where they are being traded (Digital virtual consumption). Unlike information goods like MP3s, virtual goods are rivalrous in a sense that they cannot be shared with others while simultaneously keeping them for your personal use. In more broad perspective, the definition of virtual goods encompasses a large array of virtual assets such as online avatars and their variety of accessories, digital wallpapers, movies, background and music items used for Internet games (Kim, Koh and Lee, 2009). With the increase of popularity of Social Network Communities (SNC) more scholars started turning their attention towards the consumption of virtual goods and the identification of the drivers behind that consumption. Over time. SNCs transformed into a considerable purchasing power and it didn’t take much time for gaming companies to realize the potential that virtual consumption possesses.

Ultimately, that resulted in the introduction of new form of monetization called *freemium*. Freemium business model is characterized by the delivery of a core product for free and a premium virtual content which can be purchased from the respective platform. Over the past decade this form of monetization became prevalent among internet start-ups and smartphone application developers (Kumar,2014). Social medias as LinkedIn and Skype, music and TV platforms as Spotify and Hulu or cloud service providers as Dropbox are only a few examples to name of the successful application of Freemium for the delivery of different services. Due to the low cost of initial acquisition of consumers and to the fact that ‘’free’’ is always appealing to the mass, the freemium model established itself as being more sustainable than models that rely primarily on advertising or giving free trials (Kumar,2014). Notwithstanding the undeniable success of the Freemium model, the design of a balanced service that can not only attract a big enough customer base but convert sufficient number of users to paying customers still remains a challenge that many fail to overcome. Therefore, understanding the drivers behind the consumption of premium content within the frame of freemium business model is paramount for the creation of a profitable service.

That requires a closer look at the different aspects of the consumer behavior and an investigation on the consumers' perception of the freemium product.

F2P business model is the equivalent of the freemium model adopted and applied for the purposes of the gaming industry. Often, they are even used interchangeably to describe such monetization model in games that gain revenue mainly through microtransactions. Again, the core product, the game itself, is available to anyone for no cost and additional, premium content, is offered in the game purchasable with real money. This content represents virtual items in the form of characters, virtual currencies, and tokens that can be found in the games (Lehdonvirta,2009). The roots of F2P model can be traced back to the highly profitable social games as HappyFarm, Farmville and Texas Hold'em Poker (Davidovic-Nora,2013). Their success partly derives from the rise of the social networking that spurs the interconnectedness of people using the internet platform. In the underlying concept of F2P model lies the idea that by allowing as many people as possible to play the game for free, the game developers basically facilitate the players to get a taste of the game and recognize its worth. And, since it is not about physical goods but digital ones, that worth often comes in the form of emotional reactions (Will Luton,2013).

Even though in the past F2P games were bearing the negative image of being of a low quality, the F2P model gradually overtook the gaming market. Furthermore, with the technological advancement it even became possible for the developers to create F2P games that offer a core product of a solid quality like League of legends, World of Tanks and Hawken that can stand toe to toe with their paid counterparts (K.Alha et al., 2014). That raised the bar significantly in terms of the expectations that players set for the overall service provided from the game developers. Nevertheless, games are bound to some device which make them dependable on the availability of this device. For a long time, computers and consoles were the main devices to play games on and this limitation was shortening the time of gaming that people can experienced. With the development of smartphones, that limitation was completely removed due to the fact that people basically began carrying a game platform in their pocket everywhere they go. With the exponential diffusion of smartphones, the potential of F2P model for gaining profits also jumped significantly. According to Newzoo annual report for 2018, the smartphones users for the same year hovered around three billion. That results in considerable competitiveness in the smartphone market and in mobile games turning into being the biggest segment of the games market with 573 million mobile

gamers. And, while in the past mobile games were limited only to casual genres because of the technical constraints that the mobiles devices had, today the modern smartphones are as powerful as laptops which translates into the introduction of more sophisticated gaming. Mobile games now offer more immersive and complex experience going beyond the casual gaming and that means a potential for reaching even a bigger audience (Newzoo,2018). It is expected that revenues generated from game applications alone will increase from $42,5 billion in 2016 to $106,4 billion in 2021.

With mobile gaming being set for success, game developers have to be able to profile their customers correctly if they are to be on the driving seat of the incoming mobile gaming revolution. The body of literature mostly focuses on computer and console game platforms while the mobile game environment remains understudied. This thesis shed light on the critical factors that determine the buying behavior towards freemium mobile game items. With the generated knowledge mobile game companies can gain a better holistic view of their customers, and thus to make a further step toward understanding their needs.

1.2. Research gap

Despite the undeniable success of F2P in general, the conversion rate of non-paying to paying players remains as low as 5%. In the literature, players that spend money are classified into two separate groups – the big spenders called ‘’whales’’ and ‘’minnows’’ who are players that still purchase in-game content but in significantly smaller amounts. The conditions under which F2P players transform into paying customers are still not understood completely, and thus drivers leading to certain behavior towards virtual consumption in mobile games deserve a more thorough investigation (Hamari et al., 2017).

Previous researches looked into different perspectives and different drivers that might be responsible for the manifestation of certain type of customer behaviour toward the freemium digital content. The most investigated aspects are the perception of consumer value, playfulness and flow, enjoyment of the game and satisfaction. Despite the scholars’ interest in revealing the key

motivation behind the buying behaviour in products that are practically free to use as it is in the case of F2P games, previous studies on the subject mainly take a unidimensional approach for predicting buyer behaviours. In most cases, this would involve the value that customers perceived to be received from the virtual consumption or the quality of the service and in rare cases the cultural and demographic aspects are considered. Therefore, a study that incorporates the mentioned elements is poised to further expand the knowledge about the players' behaviour toward freemium game content. On the one hand, the concept of perceived value and the concept of service quality have been tested separately but there is no frame that include service quality as a component of perceived value. Adding service quality as a predictor for the overall value that customers receive will improve significantly the robustness of the value construct. As highlighted in many past researches (Parasuraman,1997, Woodruff,1994, Sweeney,2001, Zeithaml,1988) the concept of customer perceived value bears strategic importance for the overall success of a given product or service. Nonetheless, the perception of perceived value remains ambiguous as the formation process of value can differ substantially in the varying market segments (Petrick,2004). On the other hand, there is lack of researches that take into consideration the variety of individualities in the player base and the diversity of cultural backgrounds and ethnicities while investigating the factors that influence the perception value. Lee & Wohn (2012) are among the few authors that try to investigate the influence of players' cultural dimensions on behavioural outcomes. The results of the study show that this topic needs more attention as several cultural dimensions were found to determine different usage patterns and most importantly there were evidence that cultural orientation can predict indirectly players' willingness to spend real money. However, the study limits its cultural dimensions construct to only two orientation – collectivism and invidualism, and the study sample, being comprised of only students, is not able to fully reflect the cultural diversity among the vast players' base. Hence, more developed scale of measuring cultural orientations is needed in order for managers and scholars to be able to capture the full extent to which culture can be a valuable factor in determining behavioural outcomes. Furthermore, the role of personality traits in increasing the display of some cultural orientations in a given moment and under specific circumstances has been overlooked by scholars. According to Lee & Wohn (2012) players' individualities might intertwine with their culture in a way that their combined effect predicts certain behaviours. The diversity factor is even more accentuated in the case of mobile gaming because of the increased popularity of mobile technology in general. Hence, the lack of sufficient

information on the relationship between individual characteristics such as cultural differences and the perception of value of virtual goods in mobile games leaves the feeling that there is still a lot to be discovered about the players' virtual consumption behavior.

1.3. Goals and research questions

The objective of the study is to carry on further the stream of researches that focus on the freemium business model applied in the gaming industry as the focal point of the study is the drivers behind players' intention to buy content specifically in mobile games titles. The expected expansion of the mobile game market calls for a thorough investigation on the details surrounding the formation of players' perception of value of the freemium content offered on the different platforms. Thereby, this thesis aims to fill the missing gap about the interrelation between players' individuality, the notion of perceived value and the overall willingness to pay for premium mobile game content. In view of addressing the research problem the study will tackle the following research questions:

RQ1: Does the players' perception of the perceived value of the game influence their purchase intentions and what are the main predictors of overall value?

RQ2: Do the players' personality traits have an effect on the relationship between the perceived value of the game and players' willingness to purchase, and if yes, what is it?

RQ3: RQ2: Do the players' personality traits have an effect on the relationship between the perceived value of the game and players' willingness to purchase, and if yes, what is it?

RQ4: What is the interplay of personality traits and individual cultural dimension and their combined effect on the relationship between the perceived value of the game and the willingness to purchase?

1.4. Progress of the research

The research paper opens up with the introduction of the key concepts behind the study providing justifications for the importance of the topic. In the first chapter, brief overview of the essential notions is given in order to lie the ground for the identification of the research gap that the study aims to fill.

The second chapter goes through the body of literature devoted to different elements concerning our research framework. Here, fundamental concepts and terms related to the research questions at hand are canvassed and discussed. Consequently, the ideologies and perspectives conceived as most relevant for the conditions of the study are sorted out. The chapter than continues to introduce a synopsis of the conducted researches with a similar study framework and research objectives. Based on the identified patterns the research hypotheses are formed and presented. Finally, the chapter wraps up with the illustration of the research model and with an explanation regarding the elements of the construct.

The purpose of the next chapter is to validate the selected research approach by revealing the philosophical stance of the study. Therefore, the methodology that is going to be used for linking theory with empirical research is presented. Then, the chapter goes through the process of collecting data which is followed by description of the techniques that is going to be applied for the analysis.

In the end, the results of the analysis are presented as a discussion about their meaning is developed. Also, main conclusions are made, and possible managerial implications of the research are given. Furthermore, limitations of the study are pointed out as a prelude for future research on the topic.

2. LITERATURE REVIEW

The literature review breaks down the components of our main topic, namely customers purchase behaviours in the context of mobile game industry and its antecedents. The revision of the extant literature on the topic provides justification about the choice of constructs for the measurement of the drivers behind the occurrence of certain type of customer behaviours. Each construct is reviewed as theories and previous researches regarding the subject are discussed in light of bringing awareness about the potential predictors for buying intentions.

Table 1 summarizes the quintessential research findings and concept from studies that have been used as a barebone for the theory development and hypothesis architecture in the thesis.

Table 1

Focus	Concept	Conceptualization or definition	Main results	References
Purchase motivators stemming from the game design	F2P game design and performance	The architecture of the game is conceptualized as being the reason for more concrete purchase motivators which the latent abstract constructs are unable to detect.	*Game's design* is important for players' intentions to spend money. *Artificial obstacles* (i.e. timers, spam etc.) diminish enjoyment but spurs willingness to pay while *Unobstructed play*, *social interaction*, and *economical rationale* are the most significant determinants for spending money	Hamari et al. (2017)
The influence of perceived service quality of premium services on consumer behavior	Service quality (SERVQUAL)	Modifying SERVQUAL to include only the performance indicators achieve more accurate results in terms of predictability power and measurement of consumers' behavior	*Perceived service quality* of a game affects purchase intention mainly through willingness to continue the freemium use. Only the indicator of *Reliability* of the game demonstrates significant total effect on spending for premium services when mediated by intention of continued freemium use.	Hamari et al. (2017)
How the enjoyment of the game and continuous use intentions influence purchase intentions	Perceived enjoyment and indentions for continuous use	The *enjoyment* as a concept in free-to-play games introduces a complex and not so straightforward relationship with the purchase intentions. Continuous use intentions, on the other hand, is a reliable predictor for increased purchase intentions.	*Enjoyment* of the game reduces the willingness to buy virtual goods while by setting artificial obstacles developers provide motivations for the players spend money in order to bypass them. *Attitude* toward virtual goods and the beliefs about peers' attitudes strongly increase the willingness to purchase virtual goods. Moreover, the *continued playing intention* is positively associated with purchase intentions for virtual goods.	Hamari (2017)
The value of purchasing game items and exploring factors affecting players' intention to buy them.	Adjusted *Theory of consumption* (TCV)	The modified four-items TCV scale (*Enjoyment, Monetary value, Character competency, Visual authority)* is a more appropriate measurement of the game-items value in the gaming context.	When players perceive greater value of the items and when they identify with the characters in the game they are more inclined to purchase items. Satisfaction with the game doesn't always lead to purchases.	Park, B-W and Lee, K-C (2011)

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Focus	Concept	Conceptualization or definition	Main results	References
How cultural orientations shape players' usage patterns	Collectivism and Individualism cultural orientations	Collectivism and individualism predict players' usage pattern mediated by the *expected outcomes* of the usage.	Vertical Collectivism predicts most usage patterns including buying behaviors.	Y-H. Lee & D.Y. Wohn (2012)
Factors affecting users' intention to pay for mobile game services	Perceived value and loyalty	PERVAL model for measuring perceived value is capable of measuring and predicting players' intention to play. However, the different context of the virtual world within witch people are operating leads to fluctuation in the weight of the different indicators. Thus, industries have to be tested separately for making conclusions.	Loyalty is strong determinant for paying intentions In-app purchase intentions differ between the paying and non paying groups as the latest care more about the price.	Hsiao, K-J, and Chen, C-C (2016)
Exploring the purchase intentions within the mobile apps' users	Extended expectation confirmation model (ECM)	Mobile apps' user intention is influenced by their perception of *perceived value* (performance value, value-for money, emotional value and social value), satisfaction, app rating, availability of free alternatives and habits.	Consistency between the users' expectations and what they experience determine their perception of value and satisfaction. Only *Perceived value-for-money* among the indicator of perceived value construct demonstrates effect on the purchase indentions. For current users the *satisfaction* plays important role for their purchase intentions while for the potential users social value is the most impactful one.	Hsu, C-L, and Lin, Judy, C-C (2015)
Measuring service quality of the mobile valued-added services and the relationship with value and satisfaction concepts	Service quality (SERVQUAL), perceived value and satisfaction	*Service quality* as the following for dimensions: content quality, management service, customer service and system reliability; positively influences perceived value, satisfaction and post-purchase intentions. Also, *perceived value* is regarded as predictor for users' satisfaction.	*Perceived value* and *satisfaction positively* positively influence purchase intentions. *Service quality,* however, shows no direct effect on post-purchase inentions but by having positive impact on *perceived value* and *satisfaction* it is also considered to have an indirect effect on purchase intentions	Kuo, Y-F, Wu, C-M, Deng, W-J (2009)

2.1.Consumer value

The value concept received wide attention from the academic world as many authors tried to conceptualize it in order to offer a reliable framework for managers and researches to work with. The spheres of marketing studies and customer relationship studies are the ones with the most interest in identifying the characteristics of value as the main body of literature on the subject highlights its strategic importance for companies' success (Woodruff 1997, Porter 1996, Gale 1994, Parasuraman 1997, Zeithaml 1997). According to Holbrook (1994), consumer value is '' the fundamental basis for all marketing activity". This perspective gained more popularity with the shift to marketing mix business ideology where customers' needs play a central role in companies' strategy creating process(?). Hence, the delivery of superior value leads to the establishment of competitive advantage over the competition, and thereby to bigger profits (Woodruff, 1997). Understandably, given the significance of the value concept in the modern business environment, many researchers were intrigued by the aspects of creation and delivery of superior value (Wang et al., 2004). This interest spurred the development of multiple scales and different approaches to the measurement of value. Despite the increased interest in the value concept, the task of measuring its components remains a challenge. According to Sanchez and Iniesta (2006), there are many studies on the subject, but they are all to some extent inconclusive mainly because of the complex and subjective character of the value concept. Therefore, breaking down the conceptual basis of value in order to fully understand them is what managers and research must do in light of bringing more awareness to the role that value plays in shaping consumer behaviours.

2.1.1. Perceived value

The main points of discussion arise from the two fundamental streams of studies that try to conceptualise the notion of value, namely the strategic marketing and the consumer behaviour. Therefore, the terms customer value and consumer value appeared as two different perspective of the value concept as the former generally associates with a pre-purchase evaluation and the latter with a post-purchase evaluation. However, some scholars use those terms as synonyms (Ralston

2003; Sinha and DeSarbo 1998) as it is suggested that they all refer to the consumer perception of value or otherwise said to the perceived value of a product or a service. Given that the human perception is a phenomenon that can't be limited to a certain point in time, the consumer's perceived value can occur at any stage of the purchase. The concept of perceived value is often confused with the concepts of utility, perceived value, satisfaction and perceived quality. Though, in some respects those concepts are overlapping, they are still different and distinguishable.

2.1.2. Unidimensional and Multidimensional approaches of measuring perceived value

Perceived value regarded as a unidimensional construct represents a simple trade-off between what you give and what you get in exchange. Such measurements are indicators as ''fair prices'', ''good value'' (Baker et al.2002; Sweeney et al., 1999), ''value for money'' and '' meeting quality and prices requirements'' (Grewal et al., 1998). The unidimensional construct, while not deprived of logic, is nonetheless, as Sweeney & Soutar (2001) pointed out not enough to reflect the complexity of the concept of perceived value. To improve the measurement of perceived value authors as Sweeney & Soutar and Petrick created respectively the multidimensional scale PERVAL and SERV-PERVAL. Those models treat the perceived value as a composite of several dimensions. In the case of the PERVAL model, the dimensions that constitute perceived value are emotional value, social value, functional value due to quality and functional value due to price. Similarly, the SERV-PERVAL model was designed to assess the perceived value of services. The premise of Petrick's multidimensional scale is that the unidimensional scales lack validity since they assume that consumers have a shared meaning of value, and also because there is no clear differentiation. Moreover, even though other studies showed good promises for the measurement of perceived value of tangible goods as a multidimensional scale (e.g Kantamneni and Coulson, 1996; Sweeney, Soutar and Johnson, 1998), Petrick emphasized the need for a scale specifically adapted to measuring the perceived value of a service. The SERV-PERVAL scale goes in line with the give and take perspective of the perceived value construct as the emotional, quality and reputation components are the things that consumers get from services and the monetary and non-monetary

components are what is given in return. Therefore, the emotional response was defined as the pleasure received, monetary price is the price paid as the consumer perceived it, behavioural price refers to the non-monetary price as time and effort and the reputation reflects the status of the product/service as seen by the purchaser. The results of the study show that the emotional response, quality and monetary price are the most significant predictors of perceived value.

In current thesis, the construct of perceived value goes in line with the proposed conceptual framework by Sanchez-Fernandez and Iniesta Bonillo. Based on their conceptual paperwork and on the previous literature concerning the consumer's and behavioural intentions' constructs, the following features of perceived value are suggested to best reflect its complexity. First, the consumer value is an interactive construct as it always involves two parts in the face of the consumers in the one end and the product or service in the other end. Second, the idiosyncratic and personal nature of value makes it a relative construct. Third, the value construct is preferential in the sense that it entails an evaluative judgement and perceptual because it can occur at any stage of the purchase decision process. Forth, the value construct seats at the top of the consumer's hierarchical structure since it represents a higher level of abstraction compared to the concepts of price and quality. Fifth, the value construct goes far beyond the mere cognitive perspective of a trade-off between the price paid and the quality received. As Sweeney and Soutar 2001 emphasize, this definition is way too simplistic to capture the complexity of the construct, and hence it is recommended to look from a different angle where both its cognitive and affective nature are considered.

2.2. Consumer behaviour

Generally, the most investigated factors with respect to customer behaviour are satisfaction and loyalty. It wasn't until the beginning of the 90s when the term perceived value began to spread among the scholars as before that consumer's satisfaction and loyalty were considered to be the determinants of consumer's purchase intentions of the highest value for the firms (Eggert and Ulaga, 2002). Respectively, managers need deep insight into the factors that are deemed most relevant in terms of preparing their strategy for seeking a certain type of behavioural outcomes

from the customers. With that being said, when embarking on the quest to evaluate the weight of the different factors on the overall consumer behaviour, it is paramount to identify beforehand the most important variables for the given product or service. As a result of the many studies carried out on the topic of consumer behaviour and its antecedents, there is a plethora of evidences in support of the significance of the relationship between constructs as satisfaction, quality, loyalty, value and behavioural outcomes as purchase intentions, retention rate, recommendation intentions. However, the body of literature shows mixed results regarding the overall influence of each of those factors across the different sphere of activities and the variety of products and services. Therefore, careful examination of the literature is required in order to select the most appropriate construct for the evaluation of the digital nature of the virtual items.

2.2.1. The relationship between perceived value and satisfaction, loyalty

Historically, the satisfaction construct has been given more attention and for a long time, it was considered to be a very strong predictor for behavioural intentions. There are two essential ways to look at the Satisfaction concept - transaction-specific and cumulative (Boulding et al.1993). The former is a short-term and product-specific concept that has a strongly post-purchase nature, while the latter takes into consideration the overall experience of the variety of products and services over time (Johnson and Fornell 1991). Hence, the cumulative perspective of satisfaction is considered to be the more fundamental indicator of the firm' overall performance (Anderson et al.,1994). The satisfaction construct is widely accepted to be an outcome of customers' disconfirmation process of pre-purchase expectations and post-purchase evaluation of performance indicators (Parasuraman et al.,1988), That means that the higher the discrepancy of expectation and performance, the greater the level of satisfaction or dissatisfaction that the customers experience. Consequently, if the customers are satisfied with the products or services provided by the firms, they are more likely to remain loyal (Heskett et al., 1994). Given that, the satisfaction construct is a by-product of a comparison between expectation and performance, it then follows that satisfaction is also a higher construct than the simple quality one. Furthermore, according to Kotler and Levy (1969) satisfaction is related to value. A number of authors throughout the years were

able to recognize the relationships between the satisfaction construct and the consumer value (Day and Crask 2000; Oliver 1999; Sweeney and Soutar 2001; Woodruff and Gardial 1996) but the majority of them stay firm in their statement that the two concepts while interrelated are still to be distinguished. The underlying difference is the stage at which the perceptions of value and satisfaction are formed. Consumers are expected to adopt a sense of satisfaction of a product/service after, and not before, they have the opportunity to experience it. For this reason, scholars agree upon the post-purchase character of the satisfaction construct (Caruana, Money, and Berthon 2000; Day and Crask 2000; Eggert and Ulaga 2002; Ostrom and Iacobucci 1995; Sweeney and Soutar 2001). Therefore, the investigation of those two constructs essentially provides information of different nature that bears different strategic importance. On the one hand, by investigating the satisfaction construct companies get more tactical insight about their current products and offerings, and on the other hand, the perceived value construct helps for understanding how to create value for the customers, and thus it has more long-term strategic meaning. According to Woodruff and Gardial (1996), satisfaction is just a reflection of the consumers' judgment of the value received. Naturally, if companies are capable of determining the factors that contribute to the overall creation of value for the customers, they would be able as well to influence the sense of satisfaction that the same customers are experiencing from the product/service. In addition to that, the strictly post-purchase character of satisfaction makes of it a lower-level construct in comparison with the perceived value one which can occur at any stage of the interaction between customers and products/services. For the aforementioned reasons, the perceived value construct was chosen over satisfaction as a better predictor of customer purchase intentions.

2.2.2. The relationship between perceived value and quality

Similarly, to the satisfaction construct the quality of a product is considered to have a strong connection with the notion of consumer value. Those two constructs share many common characteristics, but they have a recognizable amount of differences as well. This close relationship that exists between value and quality as two concepts that are pivotal for the marketing literature makes the use of each of them for both practitioners and scholar a confusing task. As Zeithaml (1988) notes there isn't a specific differentiation between the concepts of perceived worth and

utility, and in order to make any conclusions regarding their importance first the role that quality plays in getting value for the customers must be studied closely. Nevertheless, there are also points upon which generally scholars agree in regard to the differences between value and quality. First of all, the perceived value of a service or product is standing above the mere evaluation of their quality because of the more individualistic character of the perception of value. The superiority of the value construct is also apparent from the strictly evaluative nature of the quality construct in contrast to the trade-off ''sacrifices-benefits'' that has to happen so that the value can be determined. Therefore, value is the richer concept that has more weight on consumers choices. Nevertheless, quality is an antecedent of value and as such, it is an important component that has to be considered when an evaluation of the offered value is to be made. Therefore, to fully understand the value that players are getting from the freemium service, the quality of the game itself as a component has to be incorporated in the overall concept of value. Quality, however, is a very general notion that take into consideration different aspects depending on the given circumstances. For instance, different scales might need to be used in order to measure the quality that customers received from products and from services. The intangible character of the virtual world and the fact that game developers provide a package of different elements are the reasons to argue that what games have to offer needs to be treated as a service.

The importance of service quality has been displayed by a handful of empirical studies on the topic. In his research Ladhari (2009) reviewed and summarized the entity of researches that investigated various aspects of service quality. The common agreement among the scholars is that a superior service quality brings many benefits for the businesses, namely the possibility of achievement of such business goals as high customer retention, adoption of new customers, maintaining a healthy corporate image and spreading word-of-mouth recommendations. The concept of quality, initially goods-oriented, now predominantly shifted to mapping the perceptions of quality from a service perspective. That in return poses complications for the measurement of the construct because of the relatively more subjective nature of the perception of service compared to the perception of goods. The quality of goods can be evaluated based on aspects as their durability or physical features. Services being non-perishable and intangible cannot be measured in the same way goods do. Among the instruments developed to assess the quality of services the most recognized one is the Parasuraman's SERVQUAL scale (Ladhari, 2009). Despite the criticism towards the scale

concerning some shortcomings, it still remains the most reliable one which is widely used from both researchers and practitioners. Nevertheless, Lahdari (2009) emphasizes how important it is for the researchers to make adjustments in the scale according to the given circumstances in order to yield good results. That means that depending on the context some factors might need to be removed or altered especially across the different industries and the different services. Hamari (2017) investigates the perceived service quality of game services as for this purpose SERQUAL is applied. Hamari argues that in the given context the tangible elements of the scale (visuals, audio etc.) can be omitted as in this way better result will be produced since the game content itself is not the real focus. Also, the measurement scale was reduced to its performance items deviating from the original development of SERVQUAL which makes a comparison between pre-purchase expectation and post-purchase perceptions of the actual performance. This study will stick with the suggested SERVQUAL scale by Hamari because both pieces of research share the same context of investigating the service quality of freemium games. The scale has four items in total: Assurance, Empathy, Reliability and Responsiveness.

Assurance relates to the sense confidence and trust that the customers feel toward the service provider which in this case is the game designers. When it comes to freemium games, the assurance factor can be determined from things as players' perceptions of the trustworthiness and clarity of information, the fairness of the pricing or safety of the personal information. The lack of direct contact with the service provider makes the assurance of digital services to be greatly affected by the presence of FAQs or another source of information embedded in the game platform. The online services often include the collection of personal information which can lead to concerns regarding its safety.

Empathy corresponds to the ability of the service to tailor the experience individually for the different customers. Adapting the experience so that the customer can get the best of it is a major key for the success of the modern digital service providers (Franke, Keinz, & Steger, 2009). The customizability of the service comes in the form of various service option, flexible interface and diverse virtual items. Specifically, in the case of freemium games, customization of virtual items is highly valued as it gives players a way to build their social virtual self, and even increase their

social standing in the game environment. Customization also refers to the game experience and the number of play styles and challenges that one can experience.

Reliability

Games are electronic services and as such their technical functionality and ease of use are the main points that are taken into consideration when making an estimation regarding the service reliability. Therefore, the experience of flow and smoothness and the lack of technical problems like lag and crashes are very important for a game to be regarded as reliable.

Responsiveness of the service provider is the speed and the quality of the communication between the game designers and their players. The flow of information between players and designers is essential for the establishment of trust. The quality of the social media sites, webpages, FAQ sections can be a good indicator for the players that their service provider is carrying about the feedback and is trying to constantly improve players' experience. Problems concerning virtual items are often an issue and how game's customer support is handling situation alike is of utmost importance. Online services often experience technical issues, and responsiveness from the part of the service provider in that sense can be the frequency of patches and updates applied in view of fixing them.

Hamari (2017) investigates the effect that service quality has on players' purchase intentions. The results indicate that actual direct effect doesn't exist between the dimensions of service quality and players' intentions to pay for premium game content. Therefore, other factors might be more important when it comes to spending money on F2P games. These results come to support the decision to include in the current thesis the service quality as a dimension of the construct of perceived value, which is supposed to be directly linked to consumer intentions, rather than trying to find a direct link between the game quality and purchase intentions. Despite the lack of direct relationship between service quality and purchase intentions, Hamari's research shows that service quality is actually not to be ignored by feature studies on the topic since an indirect effect was nonetheless detected between the reliability dimension and purchase intentions. Thus, improving the reliability of the game as a service can actually lead to increased customer retention which in

return will act as a mediator for purchasing intentions. Furthermore, in another research ‘’Why do people buy virtual goods? Attitude toward virtual good purchases versus game enjoyment” (2015) Hamari revealed similar conclusions concerning the perceived enjoyment of the game and players’ attitude toward the purchasable virtual items in regard to the players’ purchase intentions. Firstly, the level of enjoyment that players experience by playing the game conversely to the conventional logic proves to have a more complicated relationship with players’ intention to spend money in the game. Several reasons related to the design of F2P games were provided for the explanation of this phenomenon. On the one hand, if the core product has already good enough quality which is accessible for everyone for no cost then there is no real incentive for players to spend money on it. On the other hand, that gap can be compensated by the means of artificial obstacles build in the game intentionally by its designers in order to, in one way or another, hinder the players’ progress. In that sense, if we suggest that holding back players to progress faster in the game reduces their enjoyment of the game than the provision of in-game content designed to accelerate the advancement will in return increase players’ willingness to buy those items. Secondly, the study’s results are in line with the idea that continued use intention is in direct relationship with players’ purchase intentions.

2.3. From Perceived value to personality traits, culture and purchase intentions

In order for game designers to be able construct their games so that they can target the desirable market segment of consumers’ having a specific set of features they first have to be confident in their ability to predict behaviours and expectations. The serious popularity of games means a vast and diverse player base of people originating from different cultural environments and possessing different sets of personality features. Thus, it becomes necessary for the gaming companies to explore the connection, if there is one, between the cultural orientations of the consumers their individual characteristics and their behaviour. This topic has been a focus for several studies that demonstrated the interrelation between cultural dimensions, personality traits and behavioural outcomes ((e.g., Griffiths, Davies, & Chappell, 2003; Steinkuehler, 2006; Williams, Yee, & Caplan, 2008, Sharma, 2010). The consensus in these studies is that, indeed, there is a correlation in place of how people perceived games, what their expectations are when approaching a game and

what they are seeking from it. Despite of the certain importance of this discovery, none of the mentioned studies seem to go beyond the broad sense of established cultural differences between separate countries. Because of this limitation, it would be incorrect to generalize the results of those studies as many other factors may contribute to the manifestation of certain type behaviour.

2.3.1. Cultural dimensions

The majority of the studies in consumer behaviour that tap into the cross-cultural aspect as a predictor of expectations and perceptions are limited to using Hofstede's or similar scales of national level of cultural factors. This raises the question of the generalizability of those studies and their validity mainly because they are deemed insufficient to assess correctly the arguably heterogeneous citizen's base. To address this issue, several studies applied scales that use individual cultural measurements (Triandis et al. 1985), the 20-item Value Survey Module, VSM 94 (Hofstede 1994), the 24-item Self-Construal Scale, SCS (Singelis 1994), the 26-item Cultural Values Scale, CVSCALE (Donthu and Yoo 1998), and the 20-item cultural dimensions scale (Furrer et al. 2000). Despite the shift from national to individual perspective, the aforementioned studies also lack validity due to their unidimensional approach for framing the separate cultural factors. One of the few studies that develop an actual multidimensional scale is the work of Triandis and Gelfand (1998). While some consider this scale to be rather reliable, other criticize it for being limited to only including the collectivism and individualism dimensions. Hence, the need for an improved rigorous scale that can measure adequately all cultural orientations with respect to their link with consumer behaviour. Piyush Sharma (2009) embarked upon this quest and developed a new 40-item scale which is a reconceptualization of the Hofstede's five-national level cultural orientations.

The scale comes together to form 10 distinctive cultural orientations: Independence and Interdependence, Power and Social inequality, Masculinity and Gender equality, Risk aversion and Ambiguity intolerance, Tradition and Prudence. The underlying concept of this study is that all 10 orientations have sub-factors, that in contrast with previous researches are characterized as co-existing features.

Independence and Interdependence

Sharma extends the prior researches that conceptualize Individualism and Collectivism as two opposite ends of the continuum as he replaced them with two negatively correlated dimensions of Independence (Individualism) and Interdependence (Collectivism). The Independence factor is associated with more independent behaviour, a developed sense of self-concept, a sense of freedom, autonomy, and personal achievements. The Collectivism factor suggests a stronger identity and an emphasis on collective achievements over personal ones.

Power and Social inequality

The Hofstede's Power Distance factor is presented as two positively correlated dimensions of Power and Social inequality. The Power factor reflects how people respond to the position of power that authority figures wield in the different organizations and the social inequality factor is what people consider normal in terms of inequalities in social ranks. High levels of Power and Social inequality factors are associated with top-down communication and pressure for the members of the society to meet certain expectations.

Risk aversion and Ambiguity avoidance

Originally, individuals with high Power distance level are seen as reluctant to take risk and determined to stick with the established norms, while those with low Power distance level are more prone to challenge the common believes and demonstrate innovative behaviors. The current study adopts the point of view that the degree of willingness to take risks and the ability to cope with ambiguities are two different things. Respectively, the Power distance dimensions is broken down into two positively correlated factors - Risk aversions and Ambiguity tolerance.

Masculinity and Gender equality

Masculine societies are normally associated with high self-ego which translates into more symbolic consumption with preference for hedonic and emotional appeals. The luxury products provide the means for individuals to differentiate themselves from others and demonstrate their success. According to Hofstede, the masculinity-femininity orientations influence the self-concept as masculine societies turn more attention to achievements and material success, while oppositely the feminine societies care more about the quality of life. Sharma reconceptualizes this understanding as two independent factors are suggested - Masculinity and Social equality. The former reflects the self-confidence and ambition that individuals possess, and the latter measure the sense of equality between men and women in terms of roles, rights and capabilities.

Tradition and Prudence

Tradition and Prudence are two positively correlated factors that represent the Hofstede's long-term orientation. On the one hand, Prudence is what Hofstede describes as long-term orientation where individuals prefer to carefully plan their steps in regards with their future. On the other hand, the Tradition factor similarly to the Hofstede's short-term orientation put focus on the well-known that brings stability, respect to tradition and opportunities for an immediate gratification instead of waiting for future benefits.

This scale is an extension to the original Hofstede's national scale of measuring cultural orientations. As the author highlights, the proposed model includes more cultural orientations adopted from previous literature on the topic. Therefore, by applying this scale managers are supposed to get more richer information about cultural differences at individual level which can be beneficial especially for companies operating in a multicultural environment.

2.3.2. Personality traits

There are two essential streams of studies that researchers have in mind when trying to pinpoint consumers' motivations to purchase. For long time, utilitarianism was regarded as the main theory that explains consumers actions to the greatest extend. The utilitarian aspects of shopping hinge on the practicability of the product, its superiority over competing products and the efficiency of shopping. Gradually, more studies started looking into the hedonic aspects of shopping adopting the assumption that purchases provide more than just utility to the customers. Hence, studies that address both the hedonic and utilitarian categories in respect to consumer's behaviour are posed to produce more holistic results. Furthermore, the concept of value perplexes researchers with its sophisticated character as there is no universal explanation of what value represents for everyone. Hence, the formation of value can be regarded as a by-product of a mix of forces that come into play before, in time of and after the purchase. Chen (2008) affirms that creation of value and preferences is related to the personality traits which in return leads to development of various purchase motivations. Following this logic, Wen-Chin and Hung-Ru (2010) investigated the effect of personality traits on the formation of hedonic and utilitarian motivations for shopping from online stores. For the research purposes, the widely used Five-Factor model of personality traits was used to prepare individual profiles of consumers with online shopping experience in Taiwan. The model consists of the following five personality traits: neuroticism, extraversion, openness to experience, agreeableness and conscientiousness.

Neuroticism as defined by R.McCrae and P.John (1992) is the individual's "tendency to experience distress, and in the cognitive and behavioural styles that follow from this tendency. People who are more neurotic tend to be also more sensitive to others opinion and it is harder for them to control their emotions. Those people are prone to make purchases on a whim.

Extraversion refers to individual's ability to communicate with other people and to establish relationships with them. Extrovert people are seen as more talkative, enthusiastic and sociable; hence they are inclined to seek excitements and gratifications.

Openness to experience is related to the range of interests that a person possesses as the higher the level of this factor more and diverse interests. Also, people who are more open to experience enjoy aesthetic impressions and demonstrate unconventional thinking.

Agreeableness is associated with the humane aspects of individuals; namely how carrying and supportive they are. Respectively, high level of agreeableness means that the individual demonstrates altruistic behaviour, he/she is easy to get along and to maintain friendly relationships with.

Conscious people are pointed as well-organised, achievement-oriented mainly because of their individualistic features. They don't act on the moment and are able to keep impulsive behaviours in check.

Wen-Chin Tsao and Hung-Ru Chang (2010) show that personality traits have a direct effect on purchase intentions and the motivations behind them. Neuroticism, extraversion and openness to experience have a link with hedonic purchase motivations. Therefore, people with those personality traits are seeking enjoyment and excitement. Particularly, higher degree of extraversion and openness to experience is associated with a higher probability of engagement in activities such entertainment and games. On the contrary, people who are more neurotic, agreeable and open to experience are more likely to look for utilitarian motivations for purchase. Hence, targeting consumers from this category means a provision of bargain prices and detailed product information.

2.4. Previous researches and hypotheses building

The literature review part provided us with information on the key metrics used in the marketing field to evaluate consumer behaviour outcomes and the factors that are pointed as most important in predicting them. The digital world presents a whole new challenge in front of game developers as the freemium business model took by storm the market and turned around the conventional understanding of customers' perception of value, and thus the overall customer behaviour. Looking back at the thesis research questions, before to start making any assumptions in that regard, it is paramount to scan carefully the existing thread of researches devoted on customer behaviour in the context of F2P games and particularly on drivers for purchasing in-game content.

The concept of value has been a focus for several researches conducted in the context of the digital consumption: mobile internet services (Hsiao 2013), mobile applications (Hsu and Lin 2015), mobile games (Hsiao and Chen, 2016), online game items (Park and Lee, 2011). The results of those studies demonstrate how subjective is the nature of the value concept and how dependant its components are on the context under which they are investigated. The majority of the studies draw on the body of literature concerning the multidimensionality of the value concept as value is normally decomposed into several subcomponents that interplay to form the overall perception of value. Despite the general understanding of value as an important indicator for purchase intentions it turns out that its components manifest different effects on the customer behaviour outcomes depending on the circumstances. For instance, while there are evidences that the emotional value and social value components have the most predictive character in terms of the customers' willingness to pay for internet service (Hsiao 2013), the same factors cannot be proved to play a role in determining customers' intention to pay for mobile applications (Hsu and Lin 2015) where the perception of value-for-money is the leading one among the applications' users. In the same vein of thoughts, Hsiao and Chen (2016) looked into the perceived value that mobile games' player experience and its relation to their intention to buy in-game content. Expectedly, the price of the virtual items emerged as the strongest predictor for purchase intentions while the playfulness of the game came at second place. At the same time, the value that players experienced from the purchase of virtual items was pointed to be most influential for their buying intentions (Park and Lee 2011). The diversity of the results concerning the influence that the different factors of value

play on customers' purchase behaviour emphasizes how important is for researchers and practitioners to avoid any generalizations in that regard. The complexity of the value concept calls for a careful selection of measurements for the specific circumstances and an independent investigation for each category of services or products. To that end, we can distinguish the following approaches to the measurement of perceived value in the digital context: transaction cost theory, theory of planned behaviour and reasoned action (Chen and Jeng,2009), technology acceptance and expectation-disconfirmation model. The majority of those studies rely on the theory of consumption values developed by Sheth, Newman, and Gross (1991) as different iterations of the original 4 values scale (quality, price, social and emotional) are proposed. However, there is a scarcity of researches on mobile games in particular as the most investigated virtual environments are social virtual worlds followed by game played on pc (Hamari, 2017). Han and Windsor (2013) and Chen and Hsiao (2013) are among the few researches that focus on mobiles games, and thus try to identify the main factors responsible for the creation of players' perception of value. In both of the study playfulness which refer to the emotions induced by the game, price of the game items and value for money were the factors that came out to demonstrate significant effect over players' purchase intentions. This fact is congruent with previous researches that suggest differences in the elements of the conception of value among the different environments. Therefore, the social value as a predictor would not be considered in the current thesis as it is not expected to play a significant role in the mobile gaming context. Interestingly, the emotional value component shows varying sings regarding the overall perception of value. Authors use the terms emotional value, enjoyment and playfulness interchangeably as they describe the same type of effective states that can come as a result of the use of products/services in the online environment. It is widely accepted that the emotional value influences positively the sense of satisfaction, loyalty and purchase intentions and many studies report results supporting that understanding, However, Hamari (2017) shows that in the case of games applying freemium business model the connection between emotions and perception of value might have a reverse effect. Hence, the negative emotions coming from intentionally embedded artificial obstacles by the developers of the game with the intention the hamper players' experience might oppositely to the conventional thinking increase the perception of value of the items and urge players to buy premium content. This observation is explained by the different design of the F2P games as some adopt highly aggressive monetization strategies. With that being said, even though researchers emphasize the significance of the relationship

between the affective emotions of the game and the perception of value, no generalization about its nature can be made since it can change along with the design of the game. Drawing on the literature, the following hypothesis was composed:

H1: Emotional response has a significant effect on the perception of value

In his original work form 2002 J. Petrick suggests that the reputation of the cruise services that are offered might mould customers' perception of their value. The adequacy of the reputation dimension in the overall concept of value was confirmed and resulted in the creation of five-dimensional scale for measuring value. Two years later, in 2004, the scale was used to test the effect of each dimensions on both current and repeated customers' purchase intentions. The study justified the strength of the image built as a valid factor that has to be considered when the quality and the value of a service is to be investigated. Free to play mobile games, which are also called apps, are predominantly offered in two main mobile platforms - IOS apps market (for apple products) and google-play-store (for android users). The significant development of the mobile technologies in the recent years led to the appearance of thousands of new apps in the mobile game markets, and that in-return made them highly saturated and extremely competitive. Consequently, it became highly important for the game developers to adopt good understanding of the dynamics of those marketplaces and how this environment, if so, can shape players' perception of the games' value and quality. In that respect, Liu and Choi (2015) took on the task to investigate the influence that game apps' reputation has on players' purchase intentions. In the mobile apps markets the reputation can be best represented by apps' ratings and users' reviews. The results of the study demonstrate the importance of those reputation indicators for the overall success of the apps. First, with good reputation apps manage to improve their visibility by climbing up to a position among the top 10 used apps which significantly increase their exposition to the users. Second, positive players' reviews can boost the image of the games and contribute for the positive attitude toward the game in general. Hsu and Lin (2015) confirm the aforementioned observation in the context of paid mobile apps as they highlight the importance for apps marketers to take the needed measures for ensuring a positive rating if they are to score good sales. Based on the carried researches concerning the quality of services' reputation in regard to the perception of value and willingness to purchase, to following hypothesis was made:

H2: Mobile games' reputation positively influences players' perception of value

Quality and price are the backbone of the value concept as they are often presented as two sides of the coin. Respectively, the perceived quality from a product or service is described to be the benefits that customers received while the price is what they have to give up in exchange. Zeithaml (1987) portrays perceived quality as "consumer's judgment about an entity's overall excellence or superiority''. With this definition in mind, the quality aspect is deemed to be integral part of the perception of value, and thus something that social researches on consumers' behaviour have always to consider. In the digital, however, often the dynamics are unpredictable and unlike the tangible goods can demonstrate unexpected results. Rezaei and Ghodsi (2014) provided evidences for the positive effect of the actual performance of MMORPGs (Massively Multiplayer Online Role-Playing games) on players' intention to buy games and to recommend them to others. Oppositely, when it comes to paying for premium content in the game, players are not willing to spend money based on the quality dimensions alone. These results show that quality is no justification for spending more than what is needed in order to obtain the core service as far as MMORPGs are concerned. In the case of paid mobile apps, Hsu and Lin (2015) were not even able to detect any significant relationship between performance indicators and attitude and intentions towards the mobile apps where the value-for-money indicator was the strongest one. At the same time, Kuo et al. (2009) conducted a study on the service quality of the mobile value-added service offered in Taiwan and revealed that the dimensions of service quality play important role in determining the level of satisfaction and the perception of value among the users. Consequently, the contradicting results about the quality aspects can be attributed to the different context of the studies. Despite of those inconsistencies, there are also persistent evidences from the gaming focused researches about the positive influence of service quality on players' continuance intentions. Hamari (2017) argue that even though the dimensions of service quality are not a direct predictor for buying behaviour, they are very relevant for retaining players via an increased willingness to continue the use of the product. Also, with spending more time playing the game, players are more likely to spend real money for premium services. Hsiao and Chen (2016) reached similar conclusions regarding the connection between mobile games' users' loyalty, mobile games' service quality aspects (labelled in the study as Access flexibility) and purchase intentions.

Therefore, based on the cues provided from the extant literature, service quality can be expected to have indirect effect on purchase intentions via latent customers' perception of the aspects of mobile games. Hence, the following hypothesis was formed:

H3: Service quality positively influences players' perception of value

In exchange of the benefits that come along with certain products or service, consumers have to make some sacrifices. This equation, as mentioned before, is the theoretical underpinning of the value concept. Most often the price that have to be paid is what is referred as sacrifices in the researches on buyers' behaviours. As Petrick (2002) specify price breaks down into two separate components: monetary price and behavioural price like time and effort required for purchasing an item. Logically, if the price is considered to be a sacrifice, lesser the perception of the sacrifice that consumers have to make better their perception of the value they can get from the product/service. It is important to note, however, that unlike the cruise services that were focus of Petrick's research, mobile gamers face fairly equal conditions in terms of time and effort to download the game or to make purchases. The reason for that is that the paying method is uniform and handled via the platforms where the games are offered. Therefore, the main form of price that will be discussed is the monetary one, Researchers adopt different perspective toward the price component and have different ways of treating it. Respectively, in the literature concerning the value of digital items price is most often represented and evaluated as : good price (Hsiao and Chen, 2016), price value (Rezaei and Ghodsi, 2014), value-for-money (Hsu and Lin, 2015; Park and Lee, 2011) and monetary price (Yoo, 2015). In a recent study from 2017 about the drivers for purchases in free-to-play games, Hamari reported that the exploratory analysis produced a factor including the following motivation for purchase: reasonable pricing, special offers, supporting a good game and investing in hobby. The common element in all of those studies is the perspective of price as the perception of the fairness of the pricing of the items and their economical nature. Furthermore, the results unanimously confirm that the price aspect of value influence significantly players' intention to spend money for in-game content. Hence, the next hypothesis is:

H4: Monetary price positively influences players' perception of value

The digitization of wide range of activities transformed different aspects of people's lives. Online shopping and gaming became widely accepted as a normal thing and that translates into their popularity among people from all ages. The increase and diversity of the online shopping consumer base means that companies have to adjust their approach of profiling their customers and try to frame the extent to which the individual personal and cultural features shape their attitude and intentions. Despite the pungent need for a better understanding of the moderating effect that consumers' individual characteristics might exert on their overall behaviour toward the entity of virtual items, there is still dearth of researches on that matter. Tsao and Chang (2010) try to address the void in the literature concerning the influence that personality traits have on online shopping behaviour. The aim of the study is to connect personality traits to the specific display of utilitarian and hedonic motivations for purchasing from online shops. Costa and McCrae's scale of big five personality traits were applied to measure individual differences in online shoppers. The different personal orientations demonstrated varying effects on the hedonic and utilitarian drivers for purchase intentions. The main conclusion that can be derived from the study is that neurotic, agreeable and open to experience people tend to look for rational reasons for purchasing online and the utilitarian aspect was the main predictor for purchase intentions in general. This observation can be explained with the fact that online shopping is often preferred over traditional brick and mortar shops because of the convenience that is offered in the form of significantly lesser amount of time and efforts needed to find and order something. However, as far as games are concerned utility-based logic is not so obviously defined as a leading criterion for players to buy in-game content because of the inherently different drivers for playing games compared to shopping. To that end, Lee and Wohn (2012) carried out a study on the two cultural dimensions of collectivism and individualism of the players of social network games and concrete motives for engaging with these games. The results showed small, but nonetheless significant effect of the vertical collectivism orientation on the behavioural outcomes such as an urge for social status and advancement in the game. It was even discovered an indirect relationship with players' willingness to spend real money for the games. These findings suggest that cultural orientations are important predictors for players' behaviour. However, none of the aforementioned studies take the perspective of the overall value experienced. Therefore, it is still unclear if and how culture and personality traits intertwine with the perception of value. Hinging on the cues concerning the

influence that the aspects of individuality have on the hedonic and utilitarian motives for purchasing online items, the following hypothesis was formed:

H5: Personality traits and cultural orientation moderate the relationships between players' perception of value and their willingness to purchase game items

2.5. Research model

According to Lin, Sher and Shin (2005) to avoid paradoxes between the conceptualization of the unidimensional and the multidimensional methods and the conceptual definition of the value construct researches have to apply *formative second-order, reflective first-order model*. Four essential points are emphasized in support of this assumption based on the taxonomy of multidimensional scales proposed by Jarvis et al. (2003). First, the causality of the construct has to begin from the value dimensions which are a mental trade-off between what is received and what is given, and to finally form the perception of value. The direction is from items to construct, and thus changes in the components are expected to modify the overall perception of value. Second, the benefit and sacrifice components while clearly distinct, are not interchangeable since they represent two opposite ends of the spectrum. Third, the correlation among the components of value can drastically vary depending on the different goods and services, different consumers' base and the availability of information on the market (Lin, Sher and Shin,2005). Consequently, even if a considerable correlation can be detected among the dimensions that is still not enough of a base to consider the construct as reflective. Fourth, the nomological network of the antecedents of the give and get indicators in the value construct differ since every individual weight differently the give and get attributes. Based on these guidelines the research model of the thesis was composed Fig.1. Petrick's SERV-PERVAL scale developed to assess the perceived value of services will be the backbone of the research model. The original scale has five dimensions but after careful consideration the behavioural price was excluded from the model as I argue that time and effort aspects are not relevant for the mobile gaming scope of the study. Therefore, the weight of the

following four latent dimensions will be evaluated on the overall perception of value received from the games: Emotional response, Reputation, Quality and Monetary price.

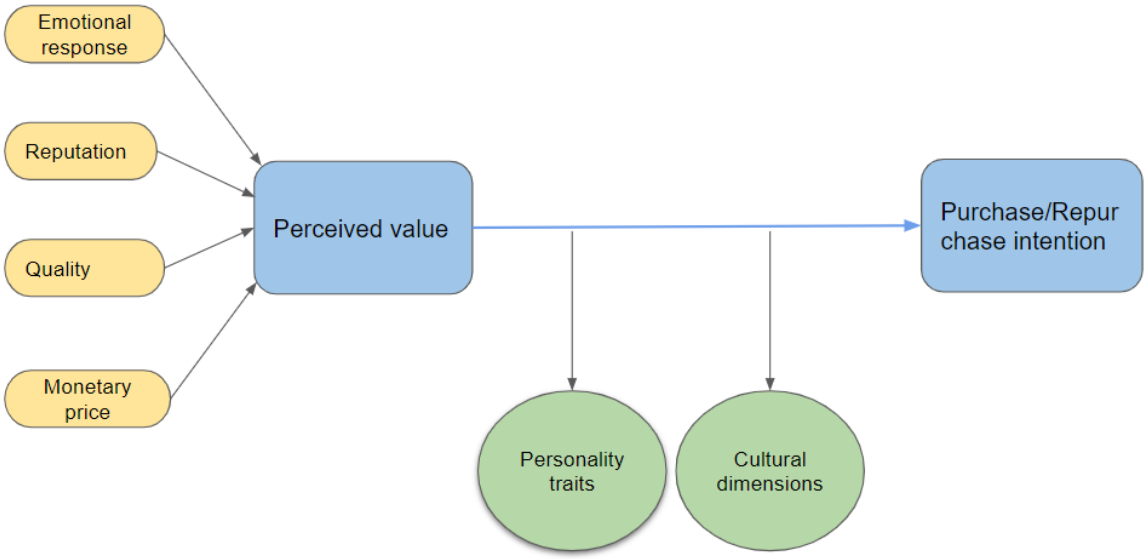


Fig.1 Research model

The second part of the model aims to determine, the single moderating effect of both personality traits and cultural dimension on the overall value and in-game items purchase intentions, and second the interplay between them. Lee and Wohn (2012) suggest a possible combined moderating effect resulting from the combination of certain sets of personal features and embedded cultural dimensions. Therefore, the nature and the strength of the combined effect might differ.

3. METHODOLOGY, DATA AND ANALYSIS METHOD

3.1. Research approach

Undertaking a study entails several philosophical and practical considerations that a researcher must have in mind before making decisions regarding the research design. According to Saunders (2009), the main purpose of every study is the generation of new knowledge and the way that researches have to be approached is dependent on how this new information is intended to be brought about to the public. Given the normal sequence of events that leads to the start of a new research, namely the conception of an idea and the development of research questions, it logically follows that first thing to be considered is the essence of the information we are seeking. The focal idea of the current thesis is to provide evidences for the existence or non-existence of a causal relationship between the latent variable perceived value, cultural and personal attributes and concrete purchase intentions. To this end, social researchers generally make distinction between two underlying research approaches - deductive and inductive. On the one hand, the use of deductive reasoning aims to produce more generalizable results, on the other hand by applying an inductive approach researchers strive to provide more in depth case-specific information. Drawing on these differences, I argue that the purpose of the thesis goes in line with the ideology of the former method mainly because it is a matter of if and not why an interrelation is at place.

The choice between deductive and inductive approach is primarily a decision regarding the way that a researcher intend to link theories with findings. Saunders (2009), however, place the origin of the research philosophy applied in the study over the matter of which research method is exactly used. Therefore, the philosophical stance that the researcher takes must be the underlying guide to the adoption of a specific study approach. Two things are considered when social researchers form the philosophical view of the study - what sort of knowledge is appropriate for the achievement of the research objectives - epistemology and what is the contextual reality in which a phenomenon is investigated - ontology. Saunders (2009) also notes that it is difficult for a research to adhere strictly to one philosophical view. As a researcher, I agree with this point of view as I am able to recognize elements in my study that are usually associated with different epistemological and

ontological assumptions. Nevertheless, I argue that the very nature of the thesis goes in line with the positivist epistemological and objectivistic ontological perspective. This true mainly because the thesis point of view is that the freemium business model possesses a distinctive set of features (objective reality) and creates an environment within which people tend to demonstrate rather consistent behaviour that can be observed (positivism), and thus quantified.

Different beliefs regarding the epistemology and ontology of a study come together to form the four research paradigms - Functionalist, Interpretative, Radical humanist and Radical structuralist (Saunders, 2009; Greener, 2008). The research paradigms help researchers to choose a reliable research method which will produce information reflecting the research objectives. Given the nature of the research, this thesis will operate within the functionalist paradigm conceptualisation and will used it to underpin the methodological approach. Thereby, a deduction is used to induce hypotheses based on the theories concerning consumers' behaviour which are tested quantitatively.

3.2. Data and its collection

The data used for the research was collected by the means of a web survey which was thoroughly put together mainly based on the knowledge and experience from previous studies. Given that the current research combines rather complex and diverse latent constructs, the first phase of creating the survey was to carefully go through the mass of studies and hand pick the most relevant questions. In the case of constructs as perceived value and perceived quality, different scales are used for their evaluation. Therefore, the approach was to look first at the context of the studies where the scales were used as what was the industry, the client's base and the products, were they intangible or tangible, for entertainment or consumption. All those factors have to be taken in consideration in order to build survey which will best capture the responses wanted from the respondents. Another consideration for the survey was to keep the time that takes to complete it as reasonable as possible. Even when the quality and the relevance of the questions are good the respondents might lose interest and patience some way through the survey if it stretches too match which in return can diminish the overall value of the gathered information. The final questionnaire

takes approximately 13 minutes for its completion which was deemed to be a reasonable time. The latter was confirmed by the test survey issued to 40 respondents in order to assess its quality before the final data collection. Three statements were prepared to estimate the ease of filling out the form: "The survey was easy to fill out", "The survey was too long", "The survey was clear". Even though, half of the respondents pointed out that the survey can be shorten a bit it has to be recognized that the width and the depth of the information that the research aims to extract requires to keep a certain volume of questions. For that reason and the fact, the more the 90% of the respondents find the survey to be easy to fill and clear enough, the yielded percent was accepted as satisfactory.

The questionnaire produced 527 valid responses which is well above the minimum of 100 recommended for the execution of a SEM analysis. The sample was additionally split into two separate groups of spenders and non-spenders with the idea of making comparison between them and potentially to lock down some of the factors that prevent people to turn into spenders. That resulted in 315 cases of people that spend money for games in one form or another and 211 cases where people hold back paying. The survey was distributed in the U.S using the online panel "Qualtrics".

Demographic characteristics

The respondents representing the two groups were further broken down by their demographic characteristics i.e age, gender, education, income and study establishment. This builds context to the study in terms of the social status of the individuals upon which later the analysis will be carried on. The overall number of female respondents was higher than the male ones scoring 60%. Surprisingly, the percent was also higher in the group of spenders (54.1%) as well as in the non-spenders (71.2%). This result hints that the sample is not influenced by the observed tendency more males to play games than females. The distribution of the age in both groups is almost identical. The age indicator was recoded forming 4 main age groups: 18-26, 27-35, 35-50 and 50+, as in both spenders and non-spenders the 35-50 age group has the more respondents with respectively 38.6% and 37%.

Bigger percent of the spenders have bachelor's degree or higher accounting for 30.7% compared to 22.8% for the non-spenders and the people with high school education or GED are the most

represented group with approximately 50%. When it comes to income, the percent in the groups of spenders (29.4%, 19.6%) and non-spenders (32%,18.5%) is close for the range of 15.000 – 35.000. 36.000 – 50.000. However, the spenders have less people with earnings less than 15.000 (13.9%) and more with earning higher than 51.000 (45.8%) compared to the non-spenders: 15.000 (19.4%) and 51.000 (29.3%).

Table 2 - Demographic characteristics

	Non-spenders		Spenders	
	N	%	N	%
Total	211	100	315	100
Gender				
Male	59	28.4	144	45.6
Female	148	71.2	171	54.1
Prefer not to answer	1	0.5	0	0
Age				
18-26	59	28	83	26.3
27-35	72	34.1	107	33.9
35-50	78	37	122	38.6
50 +	2	0.9	2	0.6
Education				
High School or GED	116	55	146	47.2
Associate Degree	44	20.9	66	20.9
Bachelor's degree	36	17.1	66	20.9
Master's degree	8	3.8	28	8.9
Doctorate Degree (MD, JD, PhD)	4	1.9	3	0.9
Income				
Less than $15,000	41	19.4	44	13.9
$15,000 - 35,000	68	32.2	93	29.4

$36,000 - 50,000	39	18.5	62	19.6
$51,000-75,000	30	14.2	53	16.8
$76,000-90,000	14	6.6	20	6.3
$91,000-119,000	10	4.7	24	7.6
More than $120,000	8	3.8	16	5.1
Study establishment				
Not studying at the time	164	77.7	201	63.6
Part-time	16	7.6	44	13.9
Full-time	29	13.7	67	21.2

The constructs were designed to use 7 points Likert scale as per the example of the researches in which they were initially used. Furthermore, this scale was recognized to be the optimal one in terms producing strong correlations and solid reliability (Symonds,1924; Lewis,1993). The descriptive analysis outlined some individual differences between the two groups regarding their perception of value and their personalities as well. First, while the *emotional value* is above the mean in both cases, spenders demonstrate slightly higher score (M=5.59) compared to the non-spenders (M=5.33). The discrepancy is way significant when evaluating the sense of safety for the personal information while playing and spending money in the game (*quality/assurance)*: spenders (M=5.32), non-spenders (M=4.61). In that regard non-spenders even demonstrate a score below the mean which is the case also for the next indicator *quality/empathy* (M=4.81). This points out to the fact that these players must feel that the games they are playing are less personalized in terms of the provided experience compared to spenders (M=5.19). In addition to that, spenders seem to be more imaginative and creative with M=5.28 to only M=4.88 for their counterpart. The following indicators, also, vary in their scores but they are similar in sense of being below the mean: *social inequality* – spenders (M=4.25), non-spenders (M=3.97); *extraversion* – spenders (M=4.91), non-spenders (M=4.45).

Table 3 - Descriptive statistics

	Spenders		Non-Spenders	
Construct	Mean	Std. dev	Mean	Std. dev
Emotional value	5.59	1.35	5.33	1.22
Price value	4.94	1.36		
Reputation value	5.54	1.19	5.41	1.15
Quality/Assurance	5.32	1.24	4.61	0.93
Quality/Reliability	5.73	1.17	5.79	1.10
Quality/Responsiveness	5.35	1.25	5.14	1.28
Quality/Empathy	5.19	1.26	4.81	1.29
Ambiguity intolerance	4.57	1.32	4.44	1.33
Tradition	5.48	1.17	5.25	1.34
Independence	5.52	1.07	5.48	1.06
Interdependence	5.53	1.07	5.32	1.23
Power	4.74	1.31	4.56	1.21
Social Inequality	4.25	1.37	3.97	1.22
Prudence	5.30	1.15	5.23	1.11
Risk Aversion	4.71	1.21	4.67	1.12
Extraversion	4.91	1.36	4.45	1.49
Openness to Experience	5.28	1.31	4.88	1.54
Agreeableness	3.47	1.91	2.84	1.69
Consciousness	5.03	1.30	4.96	1.28

3.3. Data analysis method

The current study uses structural equational modelling for testing the data and the integrity of the model. This multivariate method is also called second-generation and it has been widely used in the social sciences. The main reason for choosing second-generation method over the fist-generation techniques (multiple-regression) is the fact that SEM allow the simultaneous analysis of the model while on the other hand multiple regression requires sequential steps for probing the model. There are two main types of SEM analysis between which researchers have to choose for

carryout their analysis – covariance-based (CB-SEM) and partial least squared (PLS-SEM). There are some underlying differences that have to be taken into consideration when making the choice between them. This thesis will use PLS-SEM due to the following concerns: first, PLS-SEM tolerates the use of more vast number of indicators compared to CB-SEM; second, no consideration are made for the normality of the distribution; third, PLS-SEM handles reflective and formative models more easily resulting also in higher statistical power; forth, PLS-SEM has the edge when there is lack of prior knowledge, and, thus the emphasize of the research leans more on the exploratory side than the confirmatory one ("A primer on partial least squares structural equation modelling (PLS-SEM)", Joseph F. Hair, 2013).

4. EMPIRICAL ANALYSIS AND ITS RESULTS

In this chapter will be disclosed all the methods and approaches taken to analyse the data. The PLS path modelling is done by the SMART-PLS 3 software as the preparation of the data i.e coding the variables and the descriptive statistics were done with SPSS. The first step is to present the model evaluation, then to test it and finally to go through the results.

4.1. Measurement model assessment

The assessment of a reflective model consists of the three essential stages: evaluating the internal consistency, the convergent validity and the discriminant validity.

Internal consistency

Generally, for the evaluation of internal consistency in PLS-SEM it is taken into account the composite variability of the indicators (CR). Basically, CR measures the amount of shared variance between the indicators (Fornell & Larcker, 1981) with scores from 0 to 1. The recommended value should be above 0.7 which is the case in all indicators in our model as the range falls between 0.75

and 0.95. These values confirm that adequate consistency is achieved and thereof no redundant indicators are present. In addition, to CR
Cronbach's alpha is calculated even though it is vastly considered to be inappropriate in the context of PLS-SEM analysis due to being sensitive of the items scales and underestimating the consistency reliability. As an alternative, SMART-PLS 3 offers another indicator – rho-a, which in more recent articles has been proven to be much more accurate measure (Djikstra & Henseler, 2015). Recommended values of >0.7 are achieved for both measures Cronbach's alpha and rho_a.

Convergent validity

The convergent validity looks at the correlation between measures in the reflective construct and more specifically if they correlate positively. Reflective constructs are said to be abstract and therefore they are measured by the means of multiple indicators which are interchangeable. The idea behind this is to combine them in order to capture the abstractive character of the construct. Hence, there should be big enough shared variance between the measures in order to be converge into a reflective construct. The loadings are the first thing to check when assessing the convergent validity as the desirable values should exceed 0.5. Values below that threshold indicate that the indicator might be measuring different construct. This process of evaluating the loading is referred to as *indicator reliability*. The lowest loading of an indicator being 5.8 confirms the indicator reliability of the model's constructs. Next, the *average variance extracted* (AVE) is considered. Values above 0.5 indicates that construct explain more than half of the variance of its indicators. The AVE is calculated by summing the squared loadings and dividing them by the number of indicators. All AVE of the constructs in the model meet that requirement as the lowest one is 0.52.

Discriminant validity

The discriminant validity assesses the uniqueness of the constructs in the model. This means to provide evidences that constructs capture different phenomena. Standard techniques are to break down the *cross-loadings* and the *Fornell-Larcker criterion,* however according to Hanseler (2015) neither of the approaches is reliable when it comes to detecting discriminant validity issues. He

suggests *heterotrait-monotrait ratio* (HTMT) as a better approach. Following, Hanseler's suggestion HTMT is taken into account when assessing the discriminant validity of the constructs. Since the HTMT confidence interval of all constructs doesn't reach the threshold of 1, the discriminant validity of the construct is accepted.

The table below (table 4) summarize the measurement model assessment including the constructs that are in the core of the model.

Table 4 – Measurement model

	Items	Loadings[a]	AVE[b]	CR[c]	Rho_A[d]	CA[e]
Quality/Empathy	QE1	0.905	0.751	0.924	0.92	0.891
	QE2	0.871				
	QE2	0.839				
	QE3	0.853				
Quality/Assurance	QA1	0.900	0.785	0.916	0.909	0.866
	QA2	0.878				
	QA3	0.879				
Quality/Responsiveness	QRes1	0.843	0.76	0.927	0.896	0.895
	QRes2	0.903				
	QRes3	0.881				
	QRes4	0.859				
Quality/Reliability	QR1	0.867	0.806	0.943	0.92	0.919
	QR2	0.928				
	QR3	0.910				
	QR4	0.885				
Emotional Value	EV1	0.908	0.818	0.957	0.948	0.945
	EV2	0.910				
	EV3	0.925				
	EV4	0.907				
	EV5	0.871				

Price Value	PV1	0.880	0.769	0.93	0.934	0.902
	PV2	0.907				
	PV3	0.872				
	PV4	0.847				
Purchase Intentions	PI1	0.930	0.817	0.817	0.944	0.89
	PI2	0.937				
	PI3	0.840				
Reputation value	RV1	0.846	0.779	0.946	0.93	0.929
	RV2	0.904				
	RV3	0.904				
	RV4	0.881				
	RV5	0.877				
Ambiguity intolerance	AMB1	0.665	0.504	0.856	0.943	0.843
	AMB2	0.849				
	AMB3	0.699				
	AMB4	0.693				
	AMB5	0.797				
	AMB6	0.507				
Tradition	TRD1	0.764	0.684	0.928	0.907	0.929
	TRD2	0.844				
	TRD3	0.840				
	TRD4	0.827				
	TRD5	0.829				
	TRD6	0.854				
Independence	IND1	0.736	0.592	0.897	0.874	0.863
	IND2	0.769				
	IND3	0.769				
	IND4	0.712				
	IND5	0.808				
	IND6	0.817				
Interdependence	INT1	0.771	0.62	0.907	0.88	0.876
	INT2	0.843				
	INT3	0.801				
	INT4	0.811				
	INT5	0.806				
	INT6	0.682				

Power	POW1	0.743	0.604	0.884	0.84	0.836
	POW2	0.802				
	POW3	0.795				
	POW4	0.818				
	POW5	0.723				
Social Inequality	SIEQ1	0.788	0.575	0.87	0.823	0.811
	SIEQ2	0.841				
	SIEQ3	0.797				
	SIEQ4	0.754				
	SIEQ5	0.587				
Prudence	PRU1	0.82	0.624	0.908	0.881	0.879
	PRU2	0.84				
	PRU3	0.76				
	PRU4	0.71				
	PRU5	0.80				
	PRU6	0.81				
Risk Aversion	RSK1	0.68	0.522	0.866	0.82	0.833
	RSK2	0.71				
	RSK3	0.82				
	RSK4	0.80				
	RSK5	0.73				
	RSK6	0.56				
Extraversion	EXTR1	0.81	0.725	0.913	0.875	0.874
	EXTR2	0.86				
	EXTR3	0.88				
	EXTR4	0.85				
Openness to Experience	OTE1	0.93	0.819	0.901	0.805	0.782
	OTE2	0.89				
Agreeableness	AGR1	0.92	0.871	0.931	0.868	0.853
	AGR2	0.94				
Consciousness	CON1	0.892	0.744	0.92	0.899	0.885
	CON2	0.862				
	CON3	0.899				
	CON4	0.792				

a. All item Loadings > 0.5 indicates indicator Reliability (Hulland, 1999, p. 198)

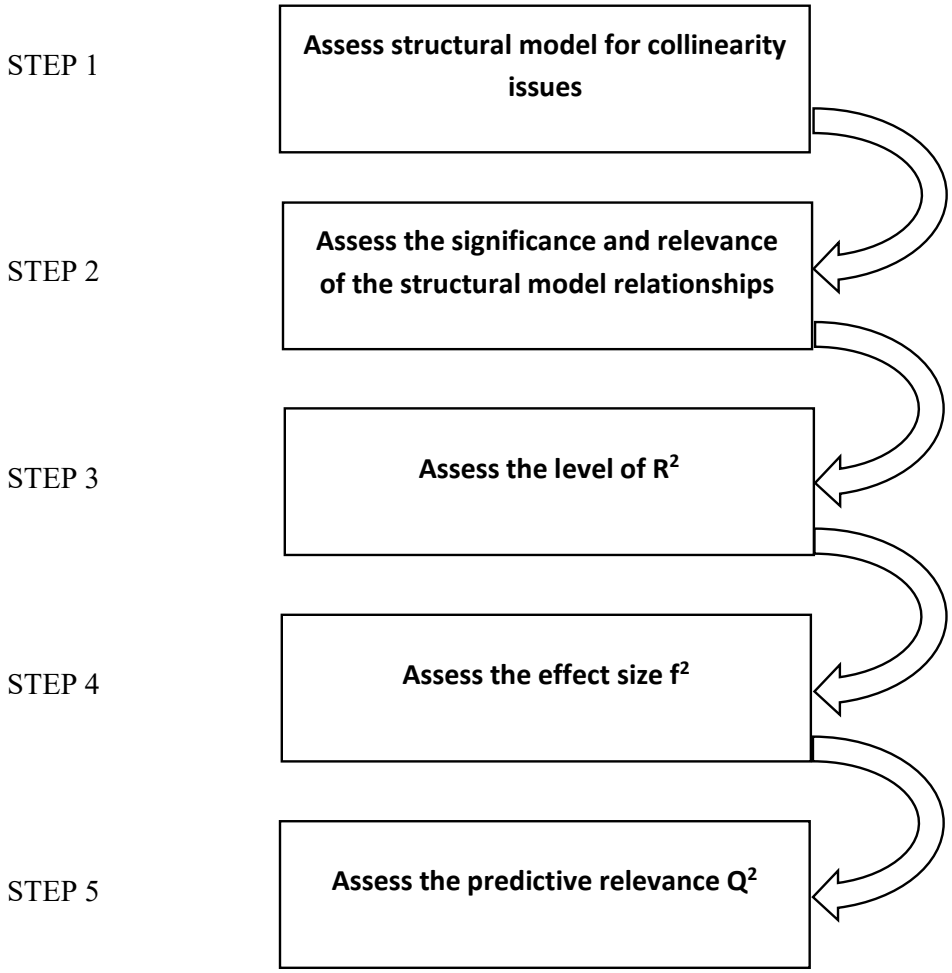
b. All Average Variance Extracted (AVE) > 0.5 as indicates Convergent reliability (Fornell & Lacker, 1981)

c. All composite reliability (CR) > 0.7 indicates Consistency (Hair et al, 2013)

d. All Cronbach's alpha > 0.7 indicates indicator Reliability (Hair et al, 2013)

4.2. Structural model assessment

After the reliability of the constructs' measures was established the analysis proceeds further with the assessment of the mode's predictive power and the existing relationship between the constructs. For that purpose, I stick to the five essential measurements outlined below.



SMART-PLS provides VIF values which are used to assess the collinearity of the constructs. The rules of thumb followed here is that the Inner VIF value should be lower than 0.5 or the construct should be revaluated, removed if need it or rebuild it into a higher construct. There were no such cases in the model so the constructs are kept as they were initially planned.

The next step suggests examining the path coefficients among all constructs for determining if they are significant and relevant. The bootstrap procedure in SMART-PLS was used for generating the t-statistics upon which the decision if a hypothesis is supported or not was made. The significance level for this study is 5% which signifies that the critical value of the two-tailed tests must be 1.96 or higher. Also, the p values are considered, as values at 0.05 provide enough basis to conclude that the given relationship is significant.
Further, to the above-mentioned t and p values, the confidence intervals are calculated to contribute for the rejection of the null hypothesis, and thus to additionally strengthen the assumption of a significant effect.

For the third step we run the PLS algorithm procedure. The R^2 give information to what extend the exogeneous variables predict the endogenous variable. The values represent the cumulative effect of all exogeneous variable and are measured between 0 and 1. Values closer to 1 indicate higher level of predictive accuracy. It is commonly accepted that values of 0.75, 0.50 and 0.25 are respectively illustrating substantial, moderate and weak latent variables.

To additionally dissect how strongly each individual construct contributes to the predictiveness of the endogenous variable (R^2 value), the effect size f^2 is manually calculated. Therefore, the constructs are taken one by one out of the model in order to observe the change in the R^2 value. The formula used is the following:
$f^2 = (R^2_{included} - R^2_{excluded})/(1-R^2_{included})$. The effect size of the exogeneous construct is considered large at 0.35, medium at 0.15 and small at 0.02(Cohen, 1988).

Similarly, to the effect size f^2 the predictive relevance of the model i.e to what degree the path model can predict the observed values is measured by Stone-Geisser Q^2 value. This value is obtained by running the *Blindfoding* procedure in SMART-PLS, and then applying the formula: $q^2 = (Q^2_{included} - Q^2_{excluded})/(1-Q^2_{included})$.

The described steps are applied for both the spenders and the non-spenders' groups and the results are summarized in **table 5** and **table 6**.

Table 5 – Model assessment Spenders

Hypothesis Relationship	Std Beta	Std Error	|t-value|^	Decision	95%CL LL
Quality Empathy ---> Overall Value	-0.013	0.068	0.162	Not Supported	-0.12
Quality Assurance ---> Overall Value	0.006	0.073	0.118	Not Supported	-0.12
Quality Responsiveness ---> Overall Value	-0.008	0.075	0.126	Not Supported	-0.14
Quality Reliability ---> Overall Value	0.147	0.096	1.630	**Almost sig**	-0.01
Emotional Value ---> Overall Value	0.338	0.085	**3.844****	**Supported**	0.21
Price Value ---> Overall Value	0.151	0.077	**1.977***	**Supported**	0.03
Reputation Value ---> Overall Value	0.120	0.109	1.067	Not Supported	-0.06
Overall value ---> Purchase Intentions	-0.069	0.046	1.386	Not Supported	-0.14
Quality Empathy ---> Purchase intentions	0.126	0.080	1.536	Not Supported	-0.01
Quality Assurance ---> Purchase intentions	0.336	0.081	**4.189****	**Supported**	0.20
Quality Responsiveness ---> Purchase intentions	0.005	0.090	0.008	Not Supported	-0.14
Quality Reliability ---> Purchase intentions	0.046	0.079	0.598	Not Supported	-0.09
Emotional Value ---> Purchase intentions	0.099	0.058	1.640	**Almost sig**	0.13
Price Value ---> Purchase intentions	0.358	0.066	**5.471****	**Supported**	0.25
Reputation Value ---> Purchase intentions	-0.077	0.067	1.148	Not Supported	-0.18

Hypothesis Relationship	f^2 *	q^2
Emotional Value ---> Overall Value	0.079	0.06881
Price Value ---> Overall Value	0.018	0.01376
Quality Assurance ---> Purchase intentions	0.098	0.05912
Price Value ---> Purchase intentions	0.124	0.07264

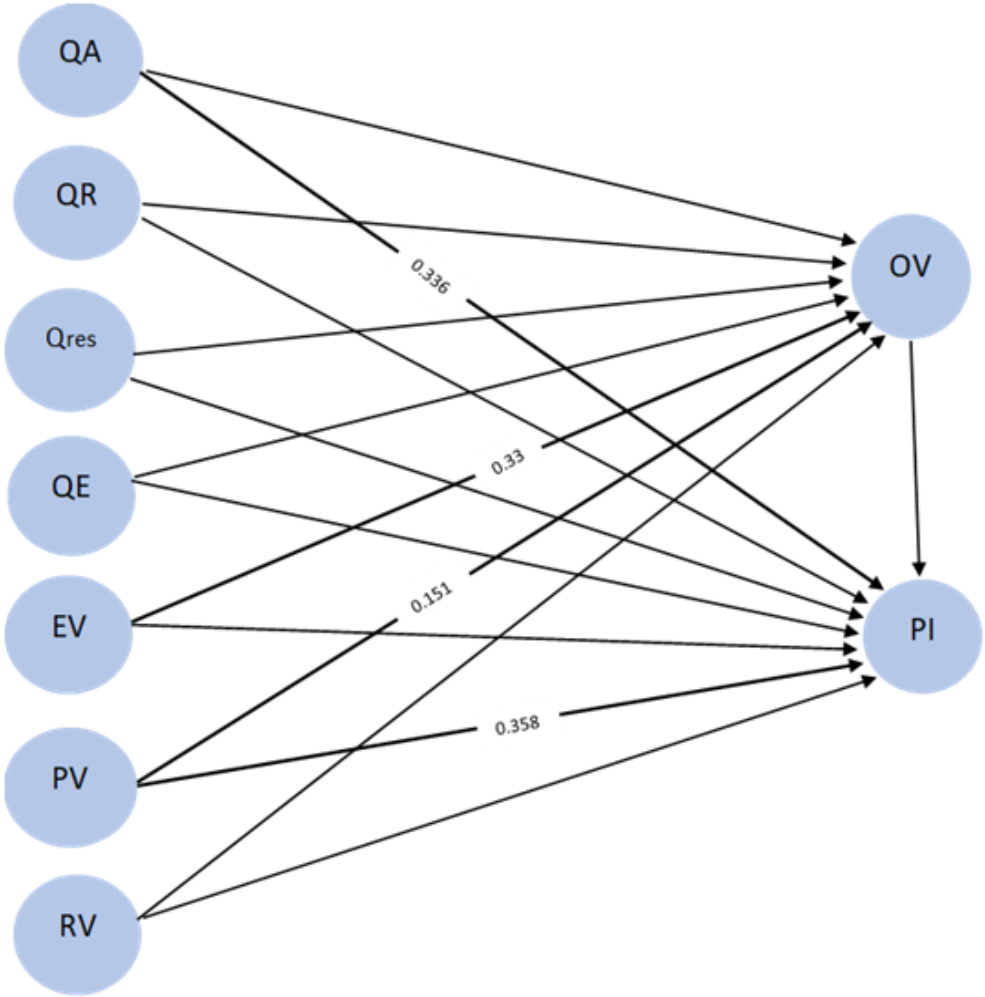
**p<0.01, *p<0.05

- Effect Size impact indicators are according Cohen (1988), f^2 values: 0.35(large), 0.15(medium), and 0.012(small)

-Predictive relevance (q^2) of Predictor Exogenous Latent Variables are according to Hanseler et al. (2009), q values: 0.35(large), 0.15(medium), and 0.012(small)

About the spenders, four path coefficients were found to be significant. Emotional value (EV) and Price value (PV) are significantly and positively influencing the overall perception of value received from the games. However, their effect size and predictive relevance are rather weak since they fall below the 0.35 threshold. Next pair of path coefficients that turned out to be significant are in direction towards the *Purchase intentions variable* (PI). Quality assurance (QA) and Price value (PV) are the two with direct effects on the players intentions to purchase now or in the future. The q^2 and f^2 in the case of QA are also weak but the overall effect size that PV has on PI can be considered medium. The structural model and the path coefficients are illustrated in figure 1.

Figure 2 – Final Model Spenders



Conversely to the spenders, the non-spenders group failed to achieve significance in the same path coefficients with exception of QA → PI relationship. Nevertheless, three new paths seem to be significant: QR → OV, QE→PI and QR→PI. The *quality/reliability* indicator is both important for

the perception of *overall value* and the *purchase intentions* for the group of players that are not converted to buyers. The effect, however, is in opposite directions for the two variables. Here, it is worth to delve into the reasons for the negative effect of QR on the PI indicator since that might turn out to be a point to improve on in order to increase the conversion into payers. In all significant relationships the effect size (f^2) and the predictive relevance (q^2) can be classified as small ones.

Table 6 – Model Assessment non-spenders

Hypothesis Relationship	Std Beta	Std Error	|t-value|^	Decision	95%CL LL	95%CL UL
Quality Empathy--->Overall Value	0.142	0.095	1.452	Not Supported	-0.01	0.30
Quality Assurance--->Overall Value	-0.038	0.104	0.611	Not Supported	-0.20	0.14
Quality Responsiveness--->Overall Value	-0.095	0.11	0.963	Not Supported	-0.27	0.09
Quality Reliability--->Overall Value	0.283	0.11	**2.563****	**Supported**	0.10	0.46
Emotional Value--->Overall Value	0.152	0.178	0.973	Not Supported	-0.14	0.44
Reputation Value--->Overall Value	0.201	0.192	1.112	Not Supported	-0.12	0.51
Overall value-->Purchase Intentions	-0.063	0.089	0.756	Not Supported	-0.21	0.08
Quality Empathy--->Purchase intentions	0.3	0.096	**3.064****	**Supported**	0.14	0.45
Quality Assurance--->Purchase intentions	0.292	0.188	**1.704***	**Supported**	-0.10	0.50
Quality Responsiveness--->Purchase intentions	-0.045	0.112	0.418	Not Supported	-0.23	0.14
Quality Reliability--->Purchase intentions	-0.281	0.112	**2.762****	**Supported**	-0.46	-0.09
Emotional Value--->Purchase intentions	0.085	0.117	0.818	Not Supported	-0.11	0.27
Reputation Value--->Purchase intentions	-0.008	0.119	0.266	Not Supported	-0.20	0.20

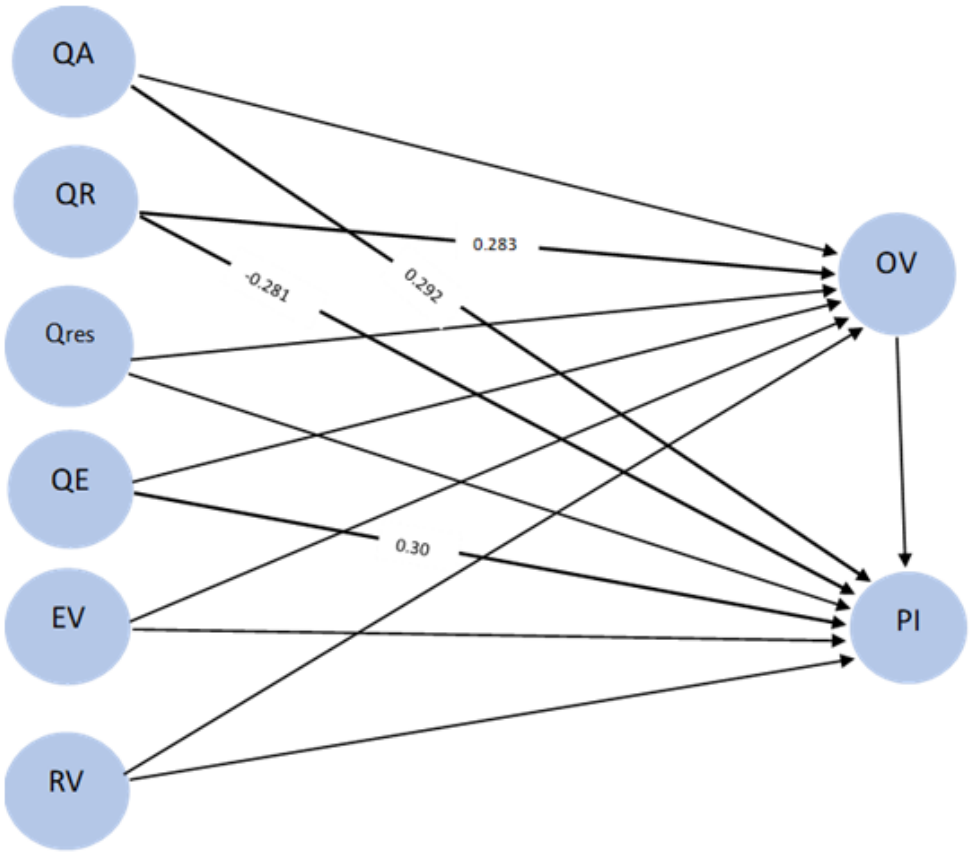
Hypothesis Relationship	f^2 *	q^2*
Quality Reliability--->Overall Value	0.047	0.035
Quality Empathy--->Purchase intentions	0.056	0.056
Quality Assurance--->Purchase intentions	0.072	0.054
Quality Reliability--->Purchase intentions	0.046	0.046

**p<0.01, *p<0.05

- Effect Size impact indicators are according Cohen (1988), f^2 values: 0.35(large), 0.15(medium), and 0.012(small)

-Predictive relevance (q^2) of Predictor Exogenous Latent Variables are according to Hanseler et al. (2009), q values: 0.35(large), 0.15(medium), and 0.012(small)

Figure 3 – Final Model Non-Spenders



4.3. Testing for moderating effects

This research's goal is not only to identify relationship between the constructs of perceived value, overall value and purchase intention but also to complete the whole picture with the possible moderating effect of cultural dimensions and personality traits. These individual characteristics are pointed out to be a good avenue and moreover a necessary one for the adoption of a more holistic view on the forces that drive purchase intentions (Hamari, 2016). This sentiment is supported by the results of the research that Wohn (2012) conducted where there were found evidences that cultural orientation might indirectly influence players' willingness to buy in-game items. The overall body of literature on the subjects is lacking researches on that matter, and therefore in a drive to contribute to filling this void, the model was test for moderation effects of cultural dimensions and personality traits constructs on the relationships between perceived value and purchase intentions.

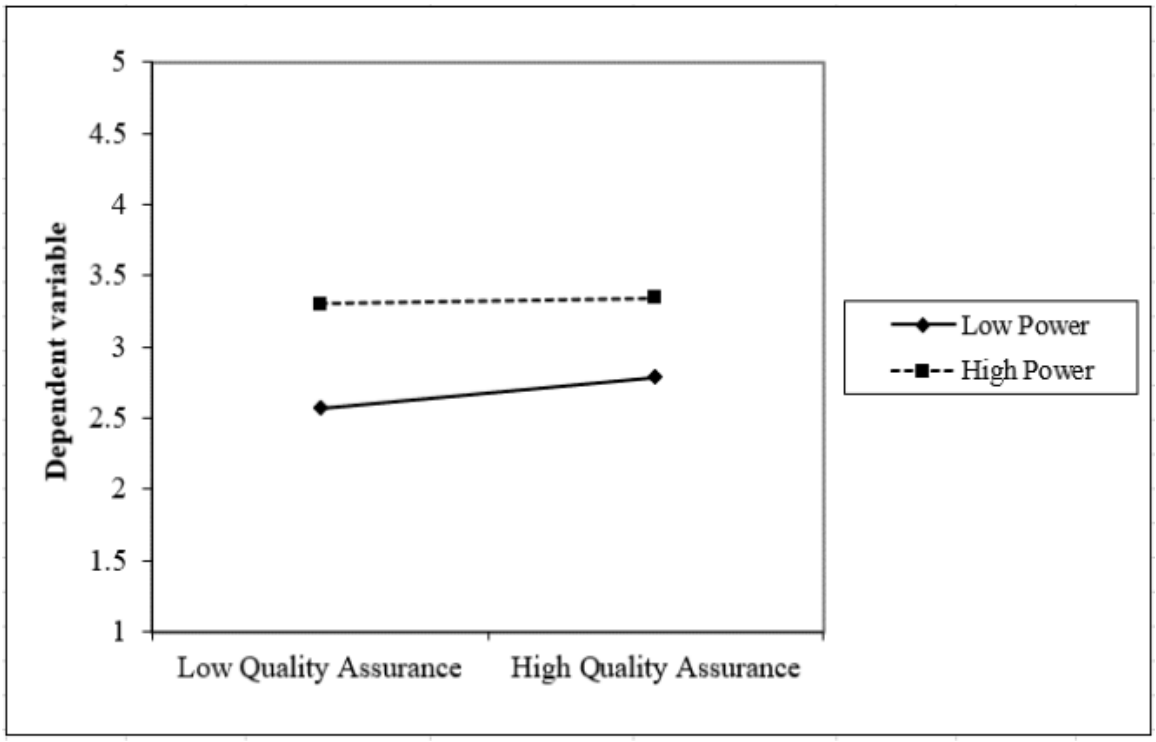
Product indicator approach was adopted for the creation of the interaction terms. That basically represents the multiplication of indicator of the exogeneous constructs with all indicators of the moderator variables. Also, following Joseph F. Hair, 2013 guidelines, the indicators were standardized in order to reduce as much as possible the collinearity that this method introduces. Then, bootstrapping procedure is initiated to test the significance of the moderating effect.

Moderation spenders

In the spenders' group, only one moderation effects were isolated from interaction term *quality assurance * power* (QA*POW). The effect is with negative impact on the otherwise positive direction of the QA → PI relationship.

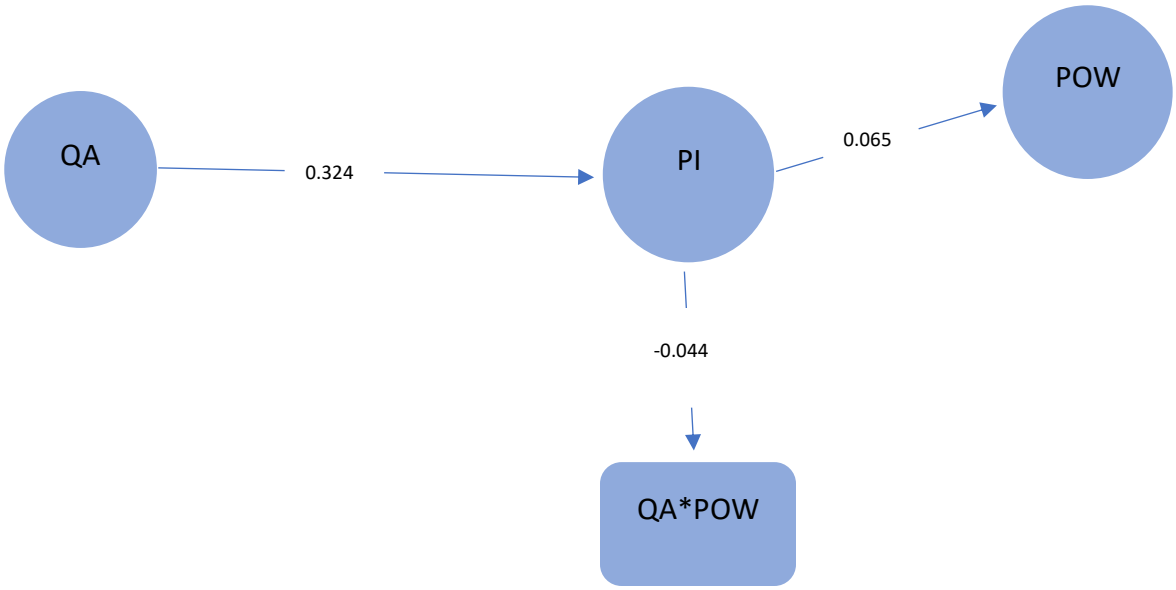
Hypothesis Relationship	Std Beta	Std Dev	|t-value|^	95%CL LL	95%CL UL
Power--->Purchase intentions	0.065	0.049	1.258	-0.01	0.149
Quality Assurance-->Purchase Intentions	0.324	0.08	4.106	0.19	0.45
Power*Quality Assurance--->Purchase intentions	-0.044	0.022	1.802*	-0.078	-0.012

Figure 4 – Moderation QA*POW



Source: http://www.jeremydawson.co.uk/slopes.htm

From **Figure 3** becomes evident that the slope for the players who score high on the cultural dimension Power is not pointing upward in contrast to the one with low Power score. That indicates a negative effect on the relationship QA → PI.

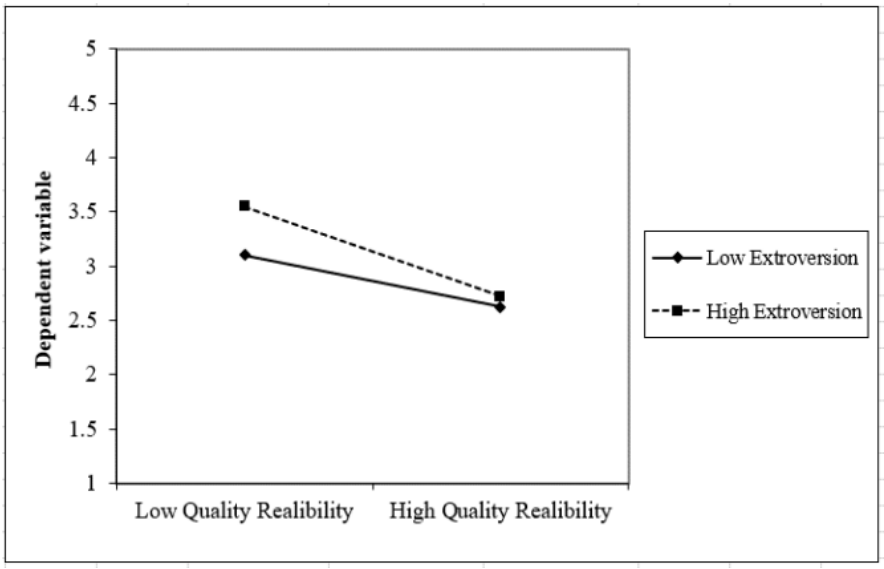


Moderation non-spenders

Withing the players which chose not spend money for in-game content, the moderation effects established are two: one personality trait (EXTR) and one cultural dimension (SIEQ). The calculated interaction terms are respectively *extraversion * quality reliability (EXTR*QR)* and *social inequality * quality reliability (SIEQ*QR)*. Both moderating effects concern the relationship between the indicator *quality reliability* and the willingness to pay for games' content. From the model assessment became clear that this indicator has negative impact on the overall *purchase intentions.* The moderation analysis provides information that can be used to further dissect why that relationship turns out to be of this sort when generally the perception of reliability of the game is associated with the overall quality of the game. The SIEQ indicator as well as the EXTR one the soften the steepness of the negative slopes when their values are on the lower end. Figure 4 and 5 demonstrates that interaction.

Hypothesis Relationship	Std Beta	Std Dev	|t-value|^	95%CL LL	95%CL UL
Social Inequality--->Purchase intentions	0.126	0.065	1.548	0.029	0.219
Quality Reliability-->Purchase Intentions	-0.246	0.123	2.268*	-0.441	-0.038
Social Inequality*Quality Reliability--->Purchase intentions	-0.095	0.123	2.268*	-0.162	-0.051
Extraversion--->Purchase intentions	0.137	0.07	1.996*	0.032	0.237
Quality Reliability-->Purchase Intentions	-0.326	0.125	2.824*	-0.517	-0.127
Extraversion*Quality Reliability--->Purchase intentions	-0.087	0.035	2.336*	-0.139	-0.033

Figure 5 – Moderation QR*EXTR



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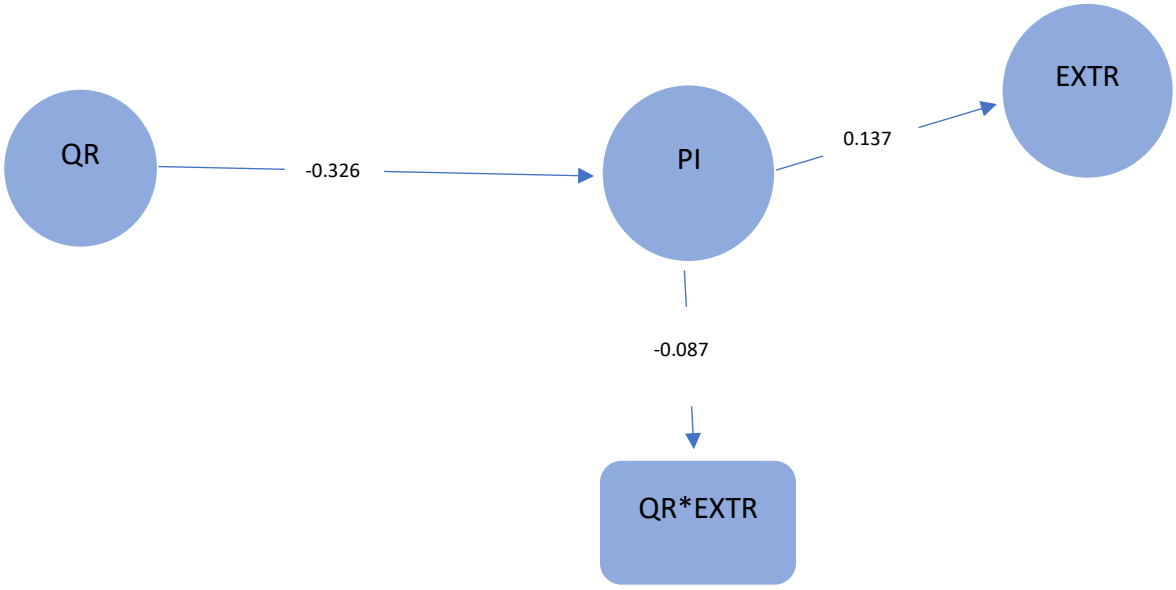
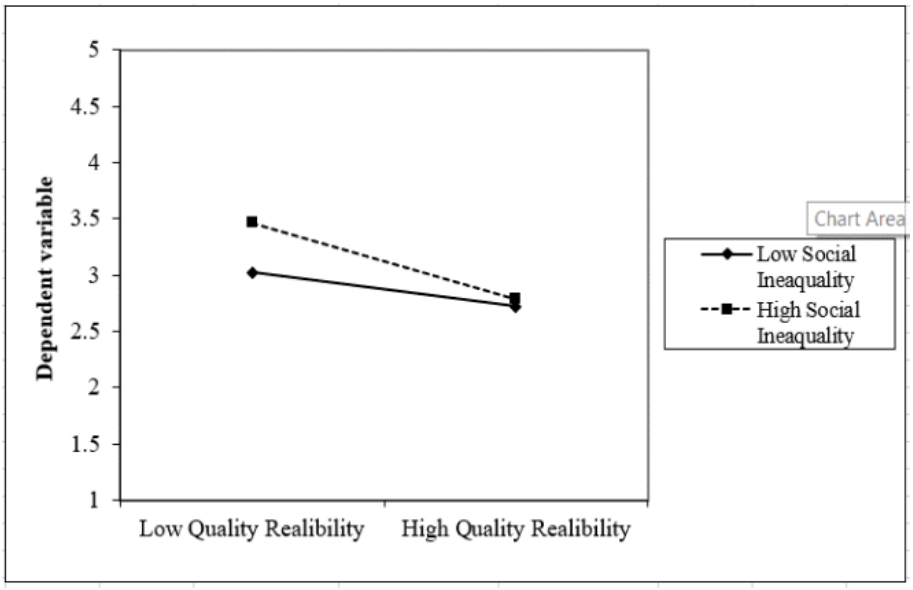


Figure 6 – Moderation QR*SIEQ



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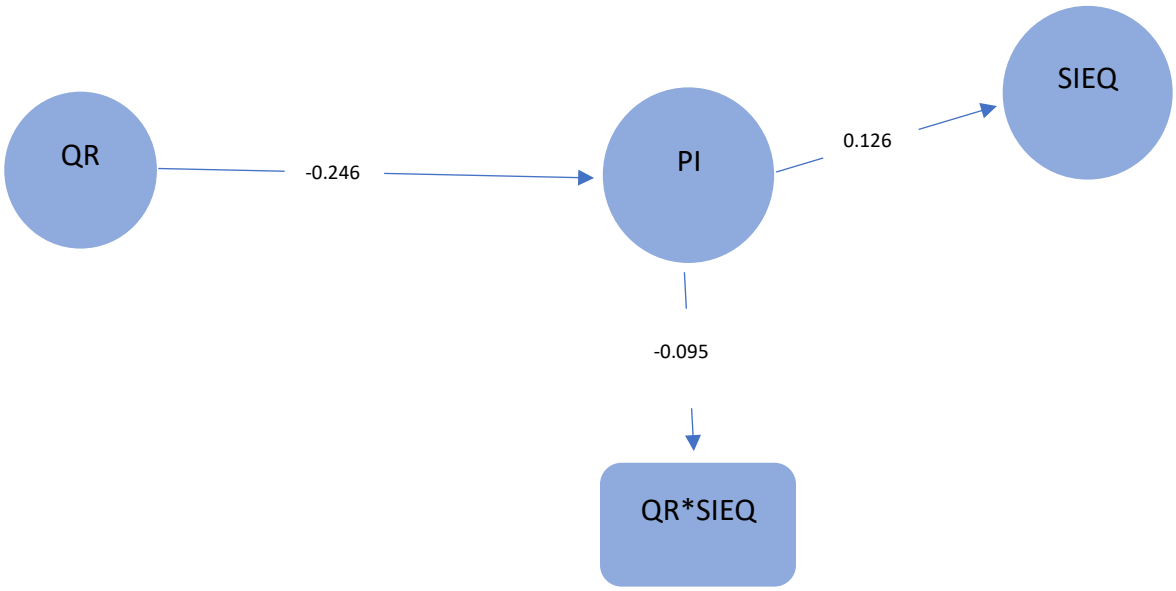


4.4. Summary of the results

The initial idea when putting together the model was to revise the understanding of the way the perception of value manipulates people's predisposition to spend real money in the virtual gaming world especially the one of the mobile games. As limited as the researches are in that area, there are several srudies that try to touch on this subject, and thus to pave the way for future researches. Nevertheless, none of the existing researches examine the indirect effect of the latent constructs as perceived value through the perception of overall value. Hence, I included the single item variable of overall value (OV) in the model in order to test the performance of this new construct in the context of the mobile gaming. The model failed to obtain significant result from the path coefficients between the *overall value* and the *purchase intentions.* Due to the lack of previous researches that incorporate such structure it is difficult the attribute this result to a single reason but what can be considered as possible factors are the complexity and width of the model as well as the single item choice of measuring the overall value indicator. Even though the *overall value* didn't demonstrate significance, indicators from the perceived value construct showed significant relationships with the OV variable as well as the intentions to buy (PI). The concept of perceived value being the heart of the model and hinging on abundant literature on that topic, was expected to come forward as a valid factor that play a role onto players' feeling of the value received from the game and their willingness to pay.

Results for spenders

In the first place, the *price value* (PI) logically is the biggest criterion for players when they assess the possibility to pay for in-game content. That comes with no surprise since many previous researches report the same results and also because it feels natural to associate the price of a product/service with the volume of sales. The effect size f^2 was determined to be medium with a value of 0.123 which is very close to the threshold of 0.15 which Cohen points as medium worthy.

That is enough to accept hypothesis 4 (H4) "Monetary price positively influences players' perception of value".

Adding the construct of Service quality adopted from the Hamari's study (2016) intended to build more complete picture of people's perception of value. Hamari makes the conclusion in his research that *service quality* has a positive effect on the players' intention the use the game but an insignificant one when it comes to their purchase intentions. However, the *reliability of the game* has an indirect positive effect on buying in-game content when it is mediated by the increased intentions for continuous use. Therefore, there were prospects to observe at least significant relationship between indicators of *service quality* and more specifically the *reliability* one with the *overall value.* Surprisingly, only *reliability* came close to score significant effect on the *overall value* with t-value of 1.63 but ultimately failed to do so. Oppositely to the sceptical expectations, however, the assurance that the game provides the players regarding the safety of their personal information when playing and paying in the game, was concluded to be significant and positive (Std.beta = 0.336; t-value = 4.189**) for their purchase intentions. This result is surprising also because nowadays the mobile games are generally considered to be secured and safe especially when they are accessed via the online stores as "Google play" and "Apple". It is worth to note that QA effect size is far from the PI one and is definitely a weak one. Based on this result for the *service quality* indicator the hypothesis "H3: Service quality positively influences players' perception of value" was rejected but new line of discussions can arise from the results for QA indicator. In that regard, a moderation effect was identified to be in place. The moderation suggests that with the level of increase of the Power orientation (POW), the already established positive effect of POW on the PI variable is brought down. This can be interpreted as the level of compliance that people demonstrated toward the figures, they estimate to be above them, can also reveal such an individual trait in them that makes them insecure about the security of the online digital environment. This though will be further developed in the discussion chapter.

The *emotional response* (EV) rightfully takes the first place in terms of weight on the overall perception of value (OV) with effect size $f^2 = 0.079$ and t-values = 3.844**. As discussed in the theory and hypothesis development parts of the thesis, games provide people with intangible product, a service, which is mostly associated with hedonic motivation for spending money. Issuing from these observations, the emotions that people feel when playing a game is with no surprise the

most important determinator of their perception of value of the game. EV→PI t-value of 1.64 is also not far from 1.96 threshold to be considered a significant relationship.

Results for non-spenders

While most of the studies that this thesis leans on for development of its theoretical background tend to focus on the paying players for their data samples and ignore the non-spenders, the current research regards this group of players as a valuable source if information. Main reason is the believe that putting the lens on the people who hold back from paying can bring out meaningful insight on why exactly that is so, and especially when there is the opportunity to make a direct comparison with their paying counterpart. Following this thought, the *emotion response (EV)* variable which is was proven to be main factor for the *overall value* (OV) among the spenders is not significant in the case of non-spenders (t-value = 0.973). The only significant relationship that emerged was the one with *quality reliability* (QR) which as mentioned failed to pan out for the spenders. Based on that, it is fair to assume that maybe the non-spenders are appreciating with higher value the overall quality of the game than the emotions coming from playing. That however on its own it is not to be taken as a sign of positive effect on the players purchase intentions. As it becomes obvious from the results, the *quality reliability* (QR) also turns out to be a significant predictor for players' purchase intentions but with negative effect (std.beta = -0.281, t-value = 2.762**). This revelation is interesting, and it is something that Hamri (2016) touched on in his research about the *service quality* role in players' use and paying intentions. Next chapter will break it down further trying to make some conclusion of it. Another contrasting thing in the non-spenders' results is the fact that three out of four indicators measuring the *service quality* construct have significant direct effects on the *purchase intentions* (PI) indicator. The *quality assurance* (QA) → PI relationship is marginally significant (t-value = 1.704) but it is brought up nevertheless since it comes in accordance with the results for the spenders. The *quality empathy* (QE) indicator demonstrate peculiar positive relationship with *purchase intentions* (PI). This indicator stands for players' point of view on the level of customization and tailor-maid experience that games provide. The analysis implies that enhancing players sense of customization provided by the game can boost their intentions to spend money in the game. Overall, for the non-spenders, only the *quality reliability*

(QR) indicator comes in support to the hypothesis that the service quality influences positively the overall perception value. The hypotheses concerning the relationship between the perceived value indicators and the OV were rejected but new avenues were discovered with the significant direct effects of *quality assurance, quality empathy* and *quality reliability* on *purchase intentions.* Furthermore, two moderating effects were found to diminish the total negative effect of *quality reliability* (QR) on players' purchase intentions. On one hand there is the personality trait *extraversion* (EXTR) that alter the negative effect QR has on PI (std.beta = -0.326) as with the interaction term EXTR*QR this effect is bring down to std.beta = -0.087. The less extroverted people appeared more prone to not be influenced by the otherwise negative effect of the reliability of the game. On the other hand, *social inequality* also has similar effect on QR→PI relationship as in this case the negative effect drops down from std.beta = -0.326 to std.beta = -0.087 when the SIEQ*QR interaction term is included in the model. This effect holds true for the people whose perception of the *social inequality* reality in the society is milder compared to those who believe strongly in it.

5. CONCLUSIONS

5.1. Discussion

The rise in popularity of the freemium business model within the gaming industry and the indisputable success that came with it, have lately been a subject of increased interest for the social sciences researches. The amount of income that this industry generates attracts multiple followers of the freemium business model. However, the recepy for building a highly profitable F-2-P games remains to certain degree ambiguous and the low percent of converted spenders testifies for that perception. Several studies embarked upon the quest to identify patterns in players' usage, their perception of value of the game and the items they buy, the relationship between the different dimensions of the quality of the game and their willingness to pay. Although, some results seem

promising, they often differ or don't provide the required depth in order to truly understand the drivers behind players intentions to pay real money for virtual goods. Scanning through the structure and the results of the studies it becomes evident that every small detail in the overall context under which they are carried, is of a significant importance for the measured effect of concepts as perceived value, perceived quality on the intention to purchase. Therefore, only using whatever results were obtained for previous researches will not be enough to make conclusions concerning the freemium model applied in the mobile gaming context. There is still lack of researches specifically targeting mobile games using F2P model and knowing the pace with which this sector grows it is paramount to invest more time in studying it. This observation served as a motivation for the development of the current thesis. It is worth to recognize that the final structure of the study is rather complex as the width of the frame includes multiple latent constructs. First, the concepts of *perceived value (SERV-PERVAL)* and *perceived quality (SERVQUAL)* were brought together to measure the *overall value* as experience by the players. Usually, in the previous studies either one of those concepts is picked to serve as a determinant for the overall value. Nevertheless, as Petrick (2002) points out the construct of perceived value can be significantly improved and complement with the inclusion of service quality as another indicator. This perception was embraced in the thesis and the expected outcome was to achieve good results regarding the predictably that those constructs provide for the overall value of the game. This expectation to some extent was met, with the model explaining 38% for the *spenders* and 31% for the *non-spenders.* The most important take away comes from the differences in the constructs' measurements between the groups. The initial plan was to take into consideration only the spenders but the good sample data that was obtained made possible to split the data into two. By doing that this research stands out from the majority of the previous researches on the subject with having the opportunity to compare *spenders* and *non-spenders* directly. Another distinctive trait of the research is the *gender* distribution of the respondents as a total of 319 female players and 203 male players. Studies examining the gaming industry are normally reporting very low percent of female respondents in their simples and that is a significant limitation since the results of those studies hinge primely on male players when in actuality their number female player is growing. In addition, the *age* distribution doesn't stick to do low or the high end and specifically the young respondents are actually not the biggest age group in the sample. Instead, the following three age groups 18-26,

27-35 and 35-50 are with very close number of representants which means that the results will be not limited to being only relevant for the youth but also for the adults.

As mentioned in the previous chapter, the *overall value* of the game wasn't proved to be a significant predictor. This might emerge a surprise because of the numerous positive results from previous studies but couple of things have to be taken into consideration: first, none of the known researches integrates the *overall value* as a separate construct; second, the single item measure wasn't tested before in this context. The ambition of the study was to incorporate the *overall value* of the game separately from the latent construct perceived value and perceived service. Despite the non-significant result, the presumption is made that future improvements in the *overall value* construct might lead to better result. Also, information about the relationship between the indicators of *perceived value* and *perceived service quality* and the *overall value* can be useful for understanding the overall sense that the players carry as to what is the estimation of the value of the game as a whole. What comes to mind when ask directly about the value of the game and the judgement made after analysing abstract latent indicators made be another avenue to explore in order to understand players' behaviour. The latent exogeneous constructs *perceived value* and *perceived service quality* was also tested in terms of their effect on the dependent variable *purchase intentions.* The results here are more promising as the path model explains 55% for the *spenders* and 19% for the *non-spenders.* Several indicators were identified as significant for players intentions to pay for the virtual goods offered in the mobile games.

Key findings

The perceived service quality indicator *assurance* was adopted from a study that examines how the effect of the perception of service quality changes when people decide to play the freemium game and when they have to make the choice to pay for in-game content or not (Hamari, 2017). In the study, *assurance* measures if the players are comfortable to pay online for in-game content, it they feel safe for their personal and bank information. Surprisingly, *assurance* turned out to be insignificant for predicting purchase intention for the player in the given sample. Oppositely to the previous results, the analysis of the current model achieved significance in the relationship between

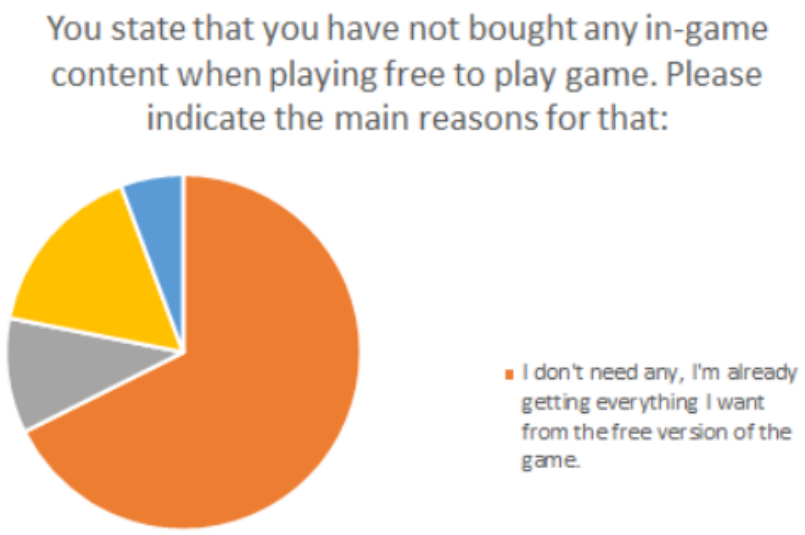
assurance and *purchase intentions* in both groups. The significance level is especially high for the *spenders* (t-value = 4.189**) what can be expected to be the case since those people are spending on a regular basis. From this result the following conclusion can be made – the players of F2P mobile games care more about the safety of the platform their using and pay more attention to the protection provided by the games. The insignificant result in the Hamari's research may come from the fact that the data sample wasn't restricted to only mobile game players. The game designers and platform providers may have to reconsider how well they inform their users about the protection set in place when personal and bank transactions are made in the online space. The selling of personal information and concerns about the bank information protection are viable topics nowadays and might prevent some people to make their first payment for online game content.

In the group of spenders, the *monetary price* indicator turns out to be valid predictor for players assessment of the *overall value* of the game as well as their willingness to purchase. The emphasis is on the relationship PV → PI as this showed the highest effect size and logically it is stated to be the biggest factor when spenders make decisions whether to buy something in the game or not. This result is not particularly surprising as in the hypotheses development part of the thesis were revealed results of several studies in favour of the expectation. All the results concerning the price value as an important factor for spending money are unanimous in their conclusion that appropriate price strategy will contribute to the demonstrated intention to purchase. In that regard, analysis of the model further solidifies this understanding for the mobile gaming context.

The emotional response that mobile games provokes in players doesn't have the same effect on their assessment for the *overall value* of the game when distinction is made between *spenders* and *non-spenders.* Emotional value, enjoyment and playfulness are terms that are used to describe the same positive effect of the gaming experience on the players that make them feel some type of joy or satisfaction of the game. The emotional response is generally associated with positive influence on people's willingness to make purchases. The body of literature mostly agrees on that matter and the expected outcome of the analysis of the model was the same. However, as Hamari hints in his study "*Why do players buy in-game content? An empirical study on concrete purchase motivations*" those otherwise viable assumptions might not be as straightforward as they appear to be given the way F2P games' design work. When it comes to the data sample of this thesis,

spenders' emotional charge received from the game undeniably influences their perception of the *overall value* (t-value EV→OV = 3.844**), while in the *non-spenders'* group that is not the case. But it is common for the groups that *emotional response* it is not a predictor for their purchase intentions. This, as Hamari explains, can be attributed to the fact that game designers intentionally embed artificial obstacles in the game experience flow and provide players with payable features for overcoming them. It is worth noting that while the relationship EV→PI it is not significant the effect is nonetheless positive so the analysis can't fully support or deny the reverse effect of enjoyment for the context of the study.

The most interesting findings of the analysis revolve around the indicators of *perceived service quality* and their relationship with the dependent variables *overall value* and *purchase intentions.* The *reliability* of the game describes the performance of the core service like the level of immersive experience that it offers. The lack of technical issues is also part of the reliability of the game. In a tradition business model, those features are regarded as important indicators for the quality of the products or services, and therefore for the sales volume. The freemium business model, however, offers a core service for free and only the premium content is for buying. Hence, more uncharacteristic and sophisticate relationship can be at place for the players of F2P mobile games. That is illustrated in the results of the analysis concerning the *non-spenders* where the *reliability* of the game while being positive and significant predictor of the *overall value*, turned out to be significant but negative predictor for *purchase intentions*. Respectively it comes the conclusion that while the *non-spenders* appreciate the quality of the performance of the game and that increases their perception of the *overall value* received, this alone is not enough to make them spend money for premium content. The reason behind that can be the different forces that lead to people starting using the free core service and to spending money. Furthermore, the various types of players can engage the game with differently intention, thus, to be looking for different aspects of a game. For instance, when a player classified as *casual* starts a game the *reliable performance* that is encountered can spur *continuous use* but also it would mean that everything the player was seeking from the game was already provided. This though is also supported by the fact 67% of the non-spenders pointed as a reason for their unwillingness to pay for in-game items the satisfying quality of the free version of the game. So, they don't feel a need for anything more than what is already free.



The negative relationship between the *reliability* of the game and players' *purchase intentions* is moderated by the personality trait *extraversion.* One of the goals of the thesis was to find out if individual and cultural difference in players have some effect on their willingness to pay. The analysis provides evidences that the level of extraversion can diminish the negative effect game's reliability. As Tsao highlights in the study "*Exploring the impact of personality traits on online shopping behaviour"* the extravert people are more socially oriented, inclined to share their experience with the others. Further, the results of the study indicate that those people are open to experience and seek enjoyment as they often engage in activities as entertainment and games. Base on that, it can be assumed that players who are extravert are also more susceptible to pay for in-game content even when the quality of the core service is good enough simply because they are seeking more interaction with other players or to increase their enjoyment. Those people, if successfully identified can be additionally targeted in order to enhance their feeling of social interaction and given attention. Also, if a big portion of the players are extraverts the game designers should consider developing the social side of its game.

Social inequality cultural orientation is the other moderating factor that distorts the negative effect that the game *reliability* has on the players' willingness to purchase. The moderating effect is similar to this of the *extraversion*. *Social inequality,* as Sharma (2009) explains in his cultural orientations scale development study, measures the acceptable level of inequality in the society as

seen from person's individual point of view. I argue that the perception of *social inequality* levels is also influenced by the degree of *extraversion* that people demonstrate. Statements "*It is difficult to interact with people from different social status than mine*" are part of the construct *social inequality* and as such it can be assumed that people with similar point of view might find it difficult to engage in social interactions which is one of the underlying traits of *extraversion.*

Single moderation effect was extracted regarding the relationship between the *spenders'* perception of *quality assurance* (i.e. the level or protection the game provides for personal and bank information) and players' *purchase intention.* As was concluded earlier in the analysis, the higher the perception of *assurance* provided the more willing spenders are to spend money. Surprisingly, this effect is neutralized when the cultural dimension *power* is added in the picture. Moreover, the effect all of sudden change direction and even demonstrates a slightly negative impact (std beta POW*QA = -0.44). The moderation analysis shows that players in the high end of *power* are not additionally motivated to purchase in game content event though they feel safe doing it. The cultural orientation *power* measures the vertical distribution of hierarchy and it is associated with tight control and a top-down type of communication (Sharma, 2010). Hollebeek suggests in his conceptualization paper "*Individual level of cultural consumer engagement styles*" (2016) that consumer's approach can vary depending on the innated individual cultural orientation. Furthermore, people considered to be less power orientated are presumably more acceptable and less critical. Conversely, high-power orientated people tend to be more critical in their judgement and expect to receive some kind of endorsements. In this line of thoughts, possible reason for the neutralizing moderating effect of *power* indicator can be that taken alone the perception of *assurance* might not provide enough grounds for untying the purse. High-power oriented people being more critical are also susceptible to have higher expectations and to be more difficult to satisfy. Therefore, game communities classified as more power-oriented have to be provided with additional incentives for spending money. To that end, *quality empathy* indicator demonstrated positive significant effect on *non-speders'* willingness to purchase. *Empathy* measures the level of customization and personalized experience provided by the game. While *non-spenders* in general don't feel need to spend for premium content when the *reliability* of the game is of high enough quality, they still can be motivated to do so by setting in place personalized game architecture. Finding of this sort can have an important implication in the management of F2P games and

specifically in the preparation of strategies for increasing the conversion from non-spenders to spenders.

5.2. Limitations and future researches

While this study attempts to make for the many of the known limitations of previous researches regarding the F2P gaming model, there are still a lot of aspects that are not covered. First, the study uses self-reported data obtained from the survey and this introduces certain level subjectivity of the information gathered and analysed. Future researches can strive to cooperated closely with game providers in order to be able to access real-usage data extracted directly from the game. This knowledge combined with the standard data from the surveys can be more valuable source of insight for what truly drives players to certain behaviours. Also, this research doesn't take account of the different game designs and monetary strategies that are applied. The different game designs can lead to specific players' behaviour and further analysis on those differences will help in identifying what this effect might be. All researches concerning freemium game business model are either using quantitative or qualitative analysis. However, I argue that the concepts of perceived value, satisfaction and perceived service quality all have a highly abstract nature and that make them really difficult to measure. Sticking to only one type of analysis hinders researches to fully grasp the depth of the information the data holds. Therefore, future researches might make an attempt to use a mix analysis of quantitative and qualitative researches. The proposes approach is to quantitively gather data through survey and after basic analysis and based on that to continue with qualitative interpretation of information obtain through interviews of small focused groups. This approach while more difficult to execute can combine the benefits of the different analysis.

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Appendices

Appendix 1 Constructs and questionnaire items

Construct	Items	source
Emotional Response (EV)	EV1: Makes me feel good EV2: Gives me pleasure EV3: Gives me a sense of joy EV4: Makes me feel delighted EV5: Gives me happiness	James F. Petrick, 2002
Monetary Price (PV)	EV1: The in-game items are worth the money EV1: The in-game items appeared to be a good bargain EV1: The in-game items are fairly priced EV1: The in-game items are reasonably priced	
Reputation (RV)	RV1: Has good reputation RV2: Is well respected RV3: Is well thought of RV4: Has status RV5: Is reputable	James F. Petrick, 2002
Quality/Assurance (QA)	QA1: I think there are unexpected risks when using money in the game QA2: I feel safe playing the game QA3: I trust the game to keep my personal information safe QA4: I feel safe about using money in the game	Hamari, 2017
Quality/Empathy (QE)	QE1: The game provides me personalized experience QE2: I think I can modify the game experience to my needs QE3: I am satisfied with the customization options in the game QE4: It is possible for me to tailor my game experience	Hamari, 2017
Quality/Reliability (QR)	QR1: The game is of good quality QR2: The game is well made QR3: I believe the game works reliably QR4: I think the game works as I expected to	Hamari, 2017
Quality/Responsiveness (Qres)	Qres1: I believe the company running the game would be responsive to my needs Qres2: In case of any problem, I think the company running the game would give me prompt service Qres3: The game's customer service will address any concerns that I have Qres4: I trust the company behind the game to offer good customer services	Hamari, 2017
Overall value (OV)	OV:Using the scale from 1 to 7, evaluate the overall value that you received from the game you play	James F. Petrick, 2002
Personality traits		Tsao & Chang, 2010
Extraversion (EXTR)	EXTR1: I am the life of the party EXTR2: I am skilled in handling social situations EXTR3: Make friends easily	

	EXTR4: Know how to captivate people
Agreeableness (AGR)	AGR1: I Insult people AGR2: I get back at others 1
Consciousness (CON)	CON1: I am always prepared CON2: I make plans and stick to them CON3: I carry out my plans CON4: I never put off till tomorrow what you can do today
Openness to Experience(OTE)	OTE1: I have unlimited creative ideas OTE2: I have ample imagination
Cultural dimensions	Lee & Wohn, 2012
Independence (IND)	IND1: I would rather depend on myself than others IND2: My personal identity, independent of others, is important to me IND3: I rely on myself most of the time, rarely on others IND4: It is important that I do my job better than others IND5: I enjoy being unique and different from others in many respects IND6: I often do 'my own thing'
Interdependence (INT)	INT1: The well-being of my group members is important for me INT2: I feel good when I cooperate with my group members INT3: It is my duty to take care of my family members, whatever it takes INT4: Family members should stick together, even if they do not agree INT5: I enjoy spending time with my group members INT6: Children must respect the decisions made by their parents
Tradition (TRD)	TRD1: I am proud of my culture TRD2: Respect for tradition is important for me TRD3: I value a strong link to my past TRD4: Traditional values are important for me TRD5: I care a lot about my family history TRD6: I always protect my family heritage
Power (POW)	POW1: I easily conform to the wishes of someone in a higher position than mine POW2: It is difficult for me to refuse a request if someone senior asks me POW3: I tend to follow orders without asking any questions POW4: I find it hard to disagree with authority figures POW5: People in higher positions have more power those in lower positions
Risk Aversion (RSK)	RSK1: I tend to avoid talking to strangers RSK2: I prefer a routine way of life to an unpredictable one full of change

	RSK3: I would not describe myself as a risk-taker RSK4: I do not like taking too many chances to avoid making a mistake RSK5: I am very cautious about how I spend my money RSK6: I am seldom the first person to try anything new
Ambiguity Intolerance (AMB)	AMB1: I find it difficult to function without clear directions and instructions AMB2: I prefer specific instructions to broad guidelines AMB3: I tend to get anxious easily when I don't know an outcome AMB4: I feel stressful when I cannot predict consequences AMB5: I feel safe when I am in my familiar surroundings AMB6: I get confused easily when dealing with complex problems
Prudence (PRD)	PRD1: I believe in planning for the long term PRD2: I work hard for success in the future PRD3: I am willing to give up today's fun for success in the future PRD4: I do not give up easily even if I do not succeed on my first attempt PRD5: I plan everything carefully PRD6: I consider many alternatives before making any decision
Social Inequality (SIEQ)	SIEQ1: A person's social status reflects his or her place in the society SIEQ2: It is important for everyone to know their rightful place in the society SIEQ3: It is difficult to interact with people from different social status than mine SIEQ4: Unequal treatment for different people is an acceptable way of life for me SIEQ5: I believe some people have an advantage over others in every society